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United States Patent	12419112
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Tanaka; Bungo

Isolation transformer

Abstract

An isolation transformer includes an insulation layer, a transformer, and a capacitor. The transformer includes first and second coils separated from each other in a thickness-wise direction of the insulation layer. The capacitor includes a first capacitor electrode and a second capacitor electrode. The insulation layer includes thin films and interlayer insulation films alternately stacked in the direction. The thin films include first and second thin films separated from each other in the direction. The interlayer insulation films include a first interlayer insulation film located next to the first thin film in the direction and a second interlayer insulation film located next to the second thin film in the direction. The first capacitor electrode is formed between the first thin film and the first interlayer insulation film. The second capacitor electrode is formed between the second thin film and the second interlayer insulation film.

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Appl. No.:	18/475261
Filed:	September 27, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240021599 A1	Jan. 18, 2024

Foreign Application Priority Data

JP	2021-055721	Mar. 29, 2021
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Related U.S. Application Data

continuation parent-doc WO PCT/JP2022/015019 20220328 PENDING child-doc US 18475261

Publication Classification

Int. Cl.: **H10D86/85** (20250101); H01L23/00 (20060101); H01L23/495 (20060101); H01L25/18 (20230101)

U.S. Cl.:

CPC **H10D86/85** (20250101); H01L23/49575 (20130101); H01L24/05 (20130101); H01L24/06 (20130101); H01L24/45 (20130101); H01L24/48 (20130101); H01L24/49 (20130101); H01L25/18 (20130101); H01L2224/05567 (20130101); H01L2224/06152 (20130101); H01L2224/0616 (20130101); H01L2224/45124 (20130101); H01L2224/45144 (20130101); H01L2224/45147 (20130101); H01L2224/48137 (20130101); H01L2224/49113 (20130101); H01L2224/49175 (20130101)

Field of Classification Search

CPC: H01F (19/04); H01F (27/00); H01L (23/49562); H01L (23/49575); H01L (24/05); H01L (24/06); H01L (24/45); H01L (24/48); H01L (24/49); H01L (25/18); H01L (2224/05567); H01L (2224/06152); H01L (2224/0616); H01L (2224/45124); H01L (2224/45144); H01L (2224/45147); H01L (2224/48137); H01L (2224/49113); H01L (2224/49175); H01L (2225/06531); H02M (1/08); H03K (17/691); H10D (1/20); H10D (1/68); H10D (86/85)

USPC: 361/270

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This present application is a bypass continuation of International Application No. PCT/JP2022/015019, filed Mar. 28, 2022, which claims priority from JP 2021-055721, filed Mar. 29, 2021, the entire contents of each are incorporated herein by reference.

BACKGROUND

- (1) The present disclosure relates to an isolation transformer.
 - (2) A known example of a gate driver that applies a gate voltage to the gate of a switching element such as a transistor is an insulated gate driver. JP 2018-78169 A describes an example of a semiconductor integrated circuit used as an insulated gate driver that includes a transformer. The transformer includes a first coil at the primary side and a second coil at the secondary side.
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Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic circuit diagram showing a first embodiment of a gate driver.
- (2) FIG. 2 is a plan view showing an internal structure of the gate driver of the first embodiment.
- (3) FIG. 3 is a perspective view showing a transformer chip included in the gate driver of the first embodiment.
- (4) FIG. 4 is a plan view of the transformer chip shown in FIG. 3.
- (5) FIG. 5 is a schematic cross-sectional view showing a first coil of the transformer chip shown in FIG. 3.
- (6) FIG. 6 is a schematic cross-sectional view showing a second coil of the transformer chip shown in FIG. 3.
- (7) FIG. 7 is a schematic cross-sectional view showing a first capacitor electrode of the transformer chip shown in FIG. 3.
- (8) FIG. 8 is a schematic cross-sectional view showing a second capacitor electrode of the transformer chip shown in FIG. 3.
- (9) FIG. 9 is a cross-sectional view of the transformer chip mounted on a low-voltage die pad taken along line 9-9 in FIG. 4.
- (10) FIG. 10 is an enlarged view showing a portion of the transformer chip shown in FIG. 9.
- (11) FIG. 11 is an enlarged view showing a portion of the transformer chip shown in FIG. 9.
- (12) FIG. 12 is a circuit diagram showing an operation of a comparative example of a gate driver.
- (13) FIG. 13 is a circuit diagram showing an operation of the gate driver in the present embodiment.
- (14) FIG. 14 is a cross-sectional view of a transformer chip mounted on a low-voltage die pad in a second embodiment of a gate driver.
- (15) FIG. 15A is an enlarged partial view showing the transformer chip of the second embodiment.
- (16) FIG. 15B is an enlarged partial view showing the transformer chip of the second embodiment.
- (17) FIG. 16A is a cross-sectional view showing a process for forming a first capacitor electrode.
- (18) FIG. 16B is a cross-sectional view showing a process for forming the first capacitor electrode.
- (19) FIG. 17 is a schematic cross-sectional view showing a first capacitor electrode of a transformer chip in a modified example.
- (20) FIG. 18 is a schematic cross-sectional view showing a first capacitor electrode of a

transformer chip in a modified example.

(21) FIG. **19** is a schematic cross-sectional view showing a first capacitor electrode of a transformer chip in a modified example.

(22) FIG. **20** is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

(23) FIG. **21** is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

(24) FIG. **22** is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

(25) FIG. **23** is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

(26) FIG. **24** is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

(27) FIG. **25** is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

(28) FIG. **26** is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

(29) FIG. **27** is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

(30) FIG. **28** is a schematic cross-sectional view of a modified example of a transformer chip showing a first coil, a second coil, a dummy coil, a first capacitor electrode, and a second capacitor electrode.

DETAILED DESCRIPTION

(31) Embodiments of a gate driver will be described below with reference to the drawings.

(32) The embodiments described below exemplify configurations and methods for embodying a technical concept and are not intended to limit the material, shape, structure, layout, dimensions, and the like of each component to those described below. Components in the drawings may be enlarged for simplicity and clarity. The dimensional proportion of a component may not be drawn to scale or may differ between drawings. In the cross-sectional views, hatching of components may be partially omitted to facilitate understanding.

First Embodiment

(33) Gate Driver

(34) A first embodiment of a gate driver **10** will be described with reference to FIGS. **1** to **11**.

(35) FIG. **1** schematically shows an example of a circuit configuration of the gate driver **10**. The gate driver **10** is configured to apply a drive voltage signal to the gate of a switching element. In an example, the gate driver **10** is used in an inverter device **500** mounted on an electric vehicle or a hybrid vehicle. The inverter device **500** includes two switching elements **501** and **502** connected in series to each other, the gate driver **10**, and an electronic control unit **503** (ECU) that controls the gate driver **10**. In an example, the switching element **501** is a high-side switching element connected to a drive power supply. The switching element **502** is a low-side switching element. Examples of the switching elements **501** and **502** include transistors such as a Si metal-oxide-semiconductor field-effect transistor (Si MOSFET), a SiC MOSFET, and an insulated gate bipolar

transistor (IGBT). The gate driver **10** of the present embodiment applies a drive voltage signal to the gate of the switching element **501**. In the description hereafter, SiC MOSFETs are used in the switching elements **501** and **502**.

(36) The gate driver **10** is provided for each of the switching elements **501** and **502** and separately drives the switching elements **501** and **502**. In the present embodiment, the gate driver **10** that drives the switching element **501** will be described for the sake of convenience.

(37) The gate driver **10** includes a low-voltage circuit **20** to which a first voltage **V1** is applied, a high-voltage circuit **30** to which a second voltage **V2** is applied, and transformers **40** arranged between the low-voltage circuit **20** and the high-voltage circuit **30**. The second voltage **V2** is higher than the first voltage **V1**. More specifically, the low-voltage circuit **20** and the high-voltage circuit **30** are connected by the transformers **40**. The first voltage **V1** and the second voltage **V2** are direct current voltages.

(38) The gate driver **10** of the present embodiment is configured, based on a control signal from the ECU **503**, to transmit a signal from the low-voltage circuit **20** to the high-voltage circuit **30** through the transformers **40** and to output a drive voltage signal from the high-voltage circuit **30**.

(39) The signal transmitted from the low-voltage circuit **20** toward the high-voltage circuit **30**, that is, a signal output from the low-voltage circuit **20**, is, for example, for driving the switching element **501**. Examples of the signal include a set signal and a reset signal. The set signal transmits a rising edge of the control signal from the ECU **503**. The reset signal transmits a falling edge of the control signal from the ECU **503**. In other words, the set signal and the reset signal are signals for generating a drive voltage signal of the switching element **501**. The set signal and the reset signal correspond to a “first signal.”

(40) More specifically, the low-voltage circuit **20** is configured to be actuated by application of the first voltage **V1**. The low-voltage circuit **20** is electrically connected to the ECU **503** and generates a set signal and a reset signal based on a control signal received from the ECU **503**. In an example, the low-voltage circuit **20** generates the set signal in response to a rising edge of the control signal and generates the reset signal in response to a falling edge of the control signal. The low-voltage circuit **20** transmits the generated set signal and reset signal toward the high-voltage circuit **30**.

(41) The high-voltage circuit **30** is configured to be actuated by application of the second voltage **V2**. The high-voltage circuit **30** is electrically connected to the gate of the switching element **501**. Based on the set signal and the reset signal received from the low-voltage circuit **20**, the high-voltage circuit **30** generates a drive voltage signal for driving the switching element **501** and applies the drive voltage signal to the gate of the switching element **501**. In other words, the high-voltage circuit **30** generates a drive voltage signal that is applied to the gate of the switching element **501** based on the first signal output from the low-voltage circuit **20**. More specifically, the high-voltage circuit **30** generates a drive voltage signal for activating the switching element **501** based on the set signal and applies the drive voltage signal to the gate of the switching element **501**. The high-voltage circuit **30** generates a drive voltage signal for deactivating the switching element **501** based on the reset signal and applies the drive voltage signal to the gate of the switching element **501**. Thus, the gate driver **10** controls the activation and deactivation of the switching element **501**.

(42) The high-voltage circuit **30** includes, for example, an R-S flip-flop circuit, into which a set signal and a reset signal are input, and a driver unit. The driver unit generates a drive voltage signal based on an output signal of the R-S flip-flop circuit. However, the high-voltage circuit **30** may be changed to any specific circuit configuration.

(43) In the gate driver **10** of the present embodiment, the low-voltage circuit **20** and the high-voltage circuit **30** are insulated from each other by the transformers **40**. More specifically, the transformers **40** restrict transmission of a direct current voltage between the low-voltage circuit **20** and the high-voltage circuit **30** while allowing transmission of various signals such as the set signal and the reset signal.

(44) Thus, a state in which the low-voltage circuit **20** and the high-voltage circuit **30** are insulated from each other refers to a state in which transmission of a direct current voltage between the low-voltage circuit **20** and the high-voltage circuit **30** is interrupted, while transmission of a signal between the low-voltage circuit **20** and the high-voltage circuit **30** is allowed.

(45) The insulation voltage of the gate driver **10** is, for example, in a range of 2500 Vrms to 7500 Vrms. In the present embodiment, the insulation voltage of the gate driver **10** is approximately 5000 Vrms. However, the insulation voltage of the gate driver **10** is not limited to these values and may be any specific numerical value.

(46) In the present embodiment, ground GND1 of the low-voltage circuit **20** and ground GND2 of the high-voltage circuit **30** are arranged independently. In the description hereafter, the potential of the ground GND1 of the low-voltage circuit **20** is referred to as a first reference potential, and the potential of the ground GND2 of the high-voltage circuit **30** is referred to as a second reference potential. In this case, the first voltage V1 is a voltage from the first reference potential, and the second voltage V2 is a voltage from the second reference potential. The first voltage V1 is, for example, in a range of 4.5 V to 5.5 V. The second voltage V2 is, for example, in a range of 9 V to 24 V.

(47) The transformers **40** will now be described in detail.

(48) The gate driver **10** of the present embodiment includes two transformers **40** and two capacitors **50** corresponding to two signals transmitted from the low-voltage circuit **20** toward the high-voltage circuit **30**. More specifically, the gate driver **10** includes a transformer **40** and a capacitor **50** that are used to transmit a set signal (SET) and a transformer **40** and a capacitor **50** that are used to transmit a reset signal (RESET). Hereinafter, for the sake of brevity, the transformer **40** and the capacitor **50** used to transmit a set signal are referred to as a “transformer **40A**” and a “capacitor **50A**.” The transformer **40** and the capacitor **50** used to transmit a reset signal are referred to as a “transformer **40B**” and a “capacitor **50B**.”

(49) The gate driver **10** includes a low-voltage signal line **21A**, which connects the low-voltage circuit **20** and the transformer **40A**, and a low-voltage signal line **21B**, which connects the low-voltage circuit **20** and the transformer **40B**. Thus, the low-voltage signal line **21A** transmits the set signal from the low-voltage circuit **20** to the transformer **40A**. The low-voltage signal line **21B** transmits the reset signal from the low-voltage circuit **20** to the transformer **40B**.

(50) The gate driver **10** includes a high-voltage signal line **31A**, which connects the transformer **40A** and the high-voltage circuit **30**, and a high-voltage signal line **31B**, which connects the transformer **40B** and the high-voltage circuit **30**. Thus, the high-voltage signal line **31A** transmits the set signal from the transformer **40A** to the high-voltage circuit **30**. The high-voltage signal line **31B** transmits the reset signal from the transformer **40B** to the high-voltage circuit **30**.

(51) The transformer **40A** electrically insulates the low-voltage circuit **20** from the high-voltage circuit **30** while transmitting the set signal from the low-voltage circuit **20** to the high-voltage circuit **30**.

(52) The transformer **40A** includes a first coil **41A** and a second coil **42A**. The first coil **41A** and the second coil **42A** are electrically insulated from each other and configured to be magnetically coupled to each other.

(53) The first coil **41A** is connected to the low-voltage circuit **20** by the low-voltage signal line **21A** and is also connected to the ground GND1 of the low-voltage circuit **20**. More specifically, the first coil **41A** includes a first end electrically connected to the low-voltage circuit **20**. The first coil **41A** is configured to receive a low voltage through the first end of the first coil **41A**. The first coil **41A** includes a second end electrically connected to the ground GND1 of the low-voltage circuit **20**. Thus, the potential of the second end of the first coil **41A** equals the first reference potential. The first reference potential is, for example, 0V.

(54) The second coil **42A** is connected to the high-voltage circuit **30** by the high-voltage signal line **31A** and is also connected to the ground GND2 of the high-voltage circuit **30**. More specifically,

the second coil **42A** includes a first end electrically connected to the high-voltage circuit **30**. The second coil **42A** is configured to receive a high voltage through the first end of the second coil **42A**. The second coil **42A** includes a second end electrically connected to the ground GND2 of the high-voltage circuit **30**. Thus, the potential of the second end of the second coil **42A** equals the second reference potential. The ground GND2 of the high-voltage circuit **30** is connected to the source of the switching element **501**. Hence, the second reference potential varies as the inverter device **500** is driven and may become, for example, greater than or equal to 600 V.

(55) The transformer **40B** electrically insulates the low-voltage circuit **20** from the high-voltage circuit **30** while transmitting the reset signal from the low-voltage circuit **20** to the high-voltage circuit **30**. The transformer **40B** includes a first coil **41B** and a second coil **42B**. The first coil **41B** and the second coil **42B** are electrically insulated from each other and configured to be magnetically coupled to each other. The connection configuration of the transformer **40B** is the same as the connection configuration of the transformer **40A** and thus will not be described in detail.

(56) The capacitor **50A** is connected to the transformer **40A**. More specifically, the capacitor **50A** is connected between the first coil **41A** and the second coil **42A** of the transformer **40A**.

(57) The capacitor **50A** includes a first capacitor electrode **51A** and a second capacitor electrode **52A**. The first capacitor electrode **51A** and the second capacitor electrode **52A** are arranged between the first coil **41A** and the second coil **42A** of the transformer **40A**. The first capacitor electrode **51A** is connected to the second end of the first coil **41A**. The second capacitor electrode **52A** is connected to the second end of the second coil **42A**. The second end of the first coil **41A** is connected to the ground GND1 of the low-voltage circuit **20**. Thus, the second end of the first coil **41A** is a ground terminal. The first capacitor electrode **51A** is connected to the ground terminal of the first coil **41A**. The second end of the second coil **42A** is connected to the ground GND2 of the high-voltage circuit **30**. Thus, the second end of the second coil **42A** is a ground terminal. Thus, the second capacitor electrode **52A** is connected to the ground terminal of the second coil **42A**.

(58) The capacitor **50B** is connected to the transformer **40B**. More specifically, the capacitor **50B** is connected between the first coil **41B** and the second coil **42B** of the transformer **40B**.

(59) The capacitor **50B** includes a first capacitor electrode **51B** and a second capacitor electrode **52B**. The first capacitor electrode **51B** and the second capacitor electrode **52B** are arranged between the first coil **41B** and the second coil **42B** of the transformer **40B**. The first capacitor electrode **51B** is connected to a ground terminal of the first coil **41B**. The second capacitor electrode **52B** is connected to a ground terminal of the second coil **42B**.

(60) FIG. 2 shows an example of a plan view showing the internal structure of the gate driver **10**. FIG. 1 shows a simplified circuit configuration of the gate driver **10**. Hence, the number of external terminals of the gate driver **10** shown in FIG. 2 is greater than the number of external terminals of the gate driver **10** shown in FIG. 1. The number of external terminals of the gate driver **10** is the number of external electrodes configured to connect the gate driver **10** to electronic components arranged outside the gate driver **10**, such as the ECU **503** or the switching element **501** (refer to FIG. 1). The number of signal lines (the number of wires W1 to W4 described later) that transmit a signal from the low-voltage circuit **20** to the high-voltage circuit **30** in the gate driver **10** shown in FIG. 2 is greater than the number of signal lines in the gate driver **10** shown in FIG. 1.

(61) As shown in FIG. 2, the gate driver **10** is a semiconductor device including multiple semiconductor chips arranged in a single package and is, for example, mounted on a circuit substrate arranged in the inverter device **500**. Each of the switching elements **501** and **502** is mounted on a mount substrate that differs from the circuit substrate. A cooling unit is attached to the mount substrate.

(62) The package type of the gate driver **10** is small outline (SO) and, in the present embodiment, is a small outline package (SOP). The gate driver **10** includes a low-voltage circuit chip **60**, a high-voltage circuit chip **70**, and a transformer chip **80**, which are semiconductor chips. The low-voltage

circuit chip **60** is mounted on a low-voltage lead frame **90**. The high-voltage circuit chip **70** is mounted on a high-voltage lead frame **100**. The chips **60**, **70**, and **80** and a portion of the lead frames **90** and **100** are encapsulated by a mold resin **110**. In the present embodiment, the transformer chip **80** corresponds to an “isolation transformer.” The transformer chip **80** and the mold resin **110** correspond to an “insulation module” that insulates the low-voltage circuit **20** from the high-voltage circuit **30**. In FIG. 2, the mold resin **110** is indicated by double-dashed lines to illustrate the internal structure of the gate driver **10**. The package type of the gate driver **10** may be changed in any manner.

(63) The mold resin **110** is formed from an electrical insulating material. An example of the resin includes a black epoxy resin. The mold resin **110** has the form of a rectangular plate having a thickness-wise direction conforming to the z-direction. The mold resin **110** includes four resin side surfaces **111** to **114**. More specifically, the mold resin **110** includes two end surfaces in the x-direction, namely, the resin side surfaces **111** and **112**, and two end surfaces in the y-direction, namely, the resin side surfaces **113** and **114**. The x-direction and the y-direction are orthogonal to the z-direction. The x-direction and the y-direction are orthogonal to each other. In the description hereafter, a plan view means a view in the z-direction.

(64) Each of the low-voltage lead frame **90** and the high-voltage lead frame **100** is formed from an electrically conductive material. The low-voltage lead frame **90** and the high-voltage lead frame **100** are formed from copper (Cu), iron (Fe), or the like. The lead frames **90** and **100** extend from the inside to the outside of the mold resin **110**.

(65) The low-voltage lead frame **90** includes a low-voltage die pad **91** arranged in the mold resin **110** and low-voltage leads **92** extending from the inside to the outside of the mold resin **110**. Each low-voltage lead **92** includes an external terminal configured to be electrically connected to an external electronic device such as the ECU **503** (refer to FIG. 1).

(66) In the present embodiment, the low-voltage circuit chip **60** and the transformer chip **80** are mounted on the low-voltage die pad **91**. In plan view, the low-voltage die pad **91** is arranged so that the center of the low-voltage die pad **91** in the y-direction is located closer in the y-direction to the resin side surface **113** than the center of the mold resin **110** is. In the present embodiment, the low-voltage die pad **91** is not exposed from the mold resin **110**. In plan view, the low-voltage die pad **91** is rectangular so that the long sides extend in the x-direction and the short sides extend in the y-direction.

(67) The low-voltage leads **92** are spaced apart from each other in the x-direction. Among the low-voltage leads **92**, the low-voltage leads **92** located at opposite ends in the x-direction are integrated with the low-voltage die pad **91**. Each low-voltage lead **92** partially projects from the resin side surface **113** toward the outside of the mold resin **110**.

(68) The high-voltage lead frame **100** includes a high-voltage die pad **101** arranged in the mold resin **110** and high-voltage leads **102** extending from the inside to the outside of the mold resin **110**. Each high-voltage lead **102** includes an external terminal configured to be electrically connected to an external electronic device such as the gate of the switching element **501** (refer to FIG. 1).

(69) The high-voltage circuit chip **70** is mounted on the high-voltage die pad **101**. In plan view, the high-voltage die pad **101** is located closer in the y-direction to the resin side surface **114** than the low-voltage die pad **91** is. In the present embodiment, the high-voltage die pad **101** is not exposed from the mold resin **110**. In plan view, the high-voltage die pad **101** is rectangular so that the long sides extend in the x-direction and the short sides extend in the y-direction.

(70) The low-voltage die pad **91** and the high-voltage die pad **101** are spaced apart from each other in the y-direction. The y-direction may also be referred to as the arrangement direction of the two die pads **91** and **101**.

(71) The dimension of the low-voltage die pad **91** and the high-voltage die pad **101** in the y-direction is set in accordance with the size and the number of semiconductor chips that are mounted. In the present embodiment, the low-voltage circuit chip **60** and the transformer chip **80**

are mounted on the low-voltage die pad **91**, and the high-voltage circuit chip **70** is mounted on the high-voltage die pad **101**. Hence, the low-voltage die pad **91** is greater than the high-voltage die pad **101** in dimension in the y-direction.

(72) The high-voltage leads **102** are spaced apart from each other in the x-direction. Among the high-voltage leads **102**, two of the high-voltage leads **102** are integrated with the high-voltage die pad **101**. Each high-voltage lead **102** partially projects from the resin side surface **114** toward the outside of the mold resin **110**.

(73) In the present embodiment, the number of the high-voltage leads **102** is the same as the number of the low-voltage leads **92**. As shown in FIG. 2, the low-voltage leads **92** and the high-voltage leads **102** are arranged in a direction (x-direction) orthogonal to the arrangement direction (y-direction) of the low-voltage die pad **91** and the high-voltage die pad **101**. The number of the high-voltage leads **102** and the number of the low-voltage leads **92** may be changed in any manner.

(74) In the present embodiment, the low-voltage die pad **91** is supported by two low-voltage leads **92** that are integrated with the low-voltage die pad **91**. The high-voltage die pad **101** is supported by two high-voltage leads **102** that are integrated with the high-voltage die pad **101**. Hence, the die pads **91** and **101** are not provided with a suspension lead exposed from the resin side surfaces **111** and **112**. This allows for an increase in the insulation distance between the low-voltage lead frame **90** and the high-voltage lead frame **100**.

(75) The low-voltage circuit chip **60**, the high-voltage circuit chip **70**, and the transformer chip **80** are spaced apart from each other in the y-direction. The low-voltage circuit chip **60**, the transformer chip **80**, and the high-voltage circuit chip **70** are arranged in this order in the y-direction from the low-voltage leads **92** toward the high-voltage leads **102**.

(76) The low-voltage circuit chip **60** includes the low-voltage circuit **20** shown in FIG. 1. In plan view, the low-voltage circuit chip **60** is rectangular and has short sides and long sides. In plan view, the low-voltage circuit chip **60** is mounted on the low-voltage die pad **91** such that the long sides extend in the x-direction and the short sides extend in the y-direction. The low-voltage circuit chip **60** includes a chip main surface **60s** and a chip back surface (not shown) facing opposite directions in the z-direction. The chip back surface of the low-voltage circuit chip **60** is bonded to the low-voltage die pad **91** by a conductive bonding material such as solder or silver (Ag) paste.

(77) First electrode pads **61**, second electrode pads **62**, and third electrode pads **63** are formed on the chip main surface **60s** of the low-voltage circuit chip **60**. The electrode pads **61** to **63** are electrically connected to the low-voltage circuit **20**.

(78) The first electrode pads **61** are located on the chip main surface **60s** between the center of the chip main surface **60s** in the y-direction and the low-voltage leads **92**. The first electrode pads **61** are arranged in the x-direction. The second electrode pads **62** are arranged on one of the opposite ends of the chip main surface **60s** in the y-direction located closer to the transformer chip **80**. The second electrode pads **62** are arranged in the x-direction. The third electrode pads **63** are arranged on opposite ends of the chip main surface **60s** in the x-direction.

(79) The high-voltage circuit chip **70** includes the high-voltage circuit **30** shown in FIG. 1. In plan view, the high-voltage circuit chip **70** is rectangular and has short sides and long sides. In plan view, the high-voltage circuit chip **70** is mounted on the high-voltage die pad **101** such that the long sides extend in the x-direction and the short sides extend in the y-direction. The high-voltage circuit chip **70** includes a chip main surface **70s** and a chip back surface (not shown) facing opposite directions in the z-direction. The chip back surface of the high-voltage circuit chip **70** is bonded to the high-voltage die pad **101** by a conductive bonding material.

(80) First electrode pads **71**, second electrode pads **72**, and third electrode pads **73** are formed on the chip main surface **70s** of the high-voltage circuit chip **70**. The electrode pads **71** to **73** are electrically connected to the high-voltage circuit **30**.

(81) The first electrode pads **71** are arranged on one of the opposite ends of the chip main surface **70s** in the y-direction located closer to the transformer chip **80**. The first electrode pads **71** are

arranged in the x-direction. The second electrode pads **72** are arranged on one of the opposite ends of the chip main surface **70s** in the y-direction located farther away from the transformer chip **80**. In other words, the second electrode pads **72** are arranged on one of the opposite ends of the chip main surface **70s** in the y-direction located closer to the high-voltage leads **102**. The second electrode pads **72** are arranged in the x-direction. The third electrode pads **73** are arranged on opposite ends of the chip main surface **70s** in the x-direction.

(82) The transformer chip **80** includes the transformers **40** (**40A**, **40B**) and the capacitors **50** (**50A**, **50B**), which are shown in FIG. **1**. In plan view, the transformer chip **80** is rectangular and has short sides and long sides. In the present embodiment, in plan view, the transformer chip **80** is mounted on the low-voltage die pad **91** such that the long sides extend in the x-direction and the short sides extend in the y-direction.

(83) The transformer chip **80** is arranged adjacent to the low-voltage circuit chip **60** in the y-direction. The transformer chip **80** is arranged closer to the high-voltage circuit chip **70** than the low-voltage circuit chip **60**. That is, the transformer chip **80** is arranged between the low-voltage circuit chip **60** and the high-voltage circuit chip **70** in the y-direction.

(84) The transformer chip **80** includes a chip main surface **80s** and a chip back surface **80r** (refer to FIG. **9**) facing in opposite directions in the z-direction. The chip back surface **80r** of the transformer chip **80** is bonded to the low-voltage die pad **91** by the conductive bonding material SD (refer to FIG. **9**).

(85) As shown in FIG. **2**, first electrode pads **81** and second electrode pads **82** are formed on the chip main surface **80s** of the transformer chip **80**. In the present embodiment, each first electrode pad **81** corresponds to a “first electrode.” Each second electrode pad **82** corresponds to a “second electrode.”

(86) The first electrode pads **81** are arranged, for example, on one of the opposite ends of the chip main surface **80s** in the y-direction located closer to the low-voltage circuit chip **60**. The first electrode pads **81** are arranged in the x-direction. The second electrode pads **82** are arranged, for example, near the center of the chip main surface **80s** in the y-direction. The second electrode pads **82** are arranged in the x-direction.

(87) As shown in FIG. **4**, in plan view, the transformers **40A** and **40B** and the capacitors **50A** and **50B** are arranged near the center of the chip main surface **80s** in the y-direction. In plan view, the second electrode pads **82** are arranged so as not to overlap with the transformers **40A** and **40B** and the capacitors **50A** and **50B**. The electrode pads **81** and **82** are electrically connected to the transformers **40A** and **40B** and the capacitors **50A** and **50B**.

(88) As shown in FIG. **2**, the lead frames **90** and **100** are located closest to each other at the low-voltage die pad **91** and the high-voltage die pad **101**. Therefore, the low-voltage die pad **91** and the high-voltage die pad **101** need to be separated from each other so that the insulation voltage of the gate driver **10** is set to a predetermined insulation voltage. Hence, in plan view, the distance between the high-voltage circuit chip **70** and the transformer chip **80** is greater than the distance between the low-voltage circuit chip **60** and the transformer chip **80**.

(89) The wires **W1** to **W4** are connected to each of the low-voltage circuit chip **60**, the transformer chip **80**, and the high-voltage circuit chip **70**. Each of the wires **W1** to **W4** is a bonding wire formed by a wire bonder and is, for example, formed from a material including gold (Au), aluminum (Al), Cu, or the like.

(90) The low-voltage circuit chip **60** is electrically connected to the low-voltage lead frame **90** by the wires **W1**. More specifically, the first electrode pads **61** and the third electrode pads **63** of the low-voltage circuit chip **60** are connected to the low-voltage leads **92** by the wires **W1**. The third electrode pads **63** of the low-voltage circuit chip **60** are connected by the wires **W1** to the two low-voltage leads **92** integrated with the low-voltage die pad **91**. Thus, the low-voltage circuit **20** is electrically connected to the low-voltage leads **92** (external electrodes of the gate driver **10** that are electrically connected to the ECU **503**). In the present embodiment, the two low-voltage leads **92**

integrated with the low-voltage die pad **91** include a ground terminal, and the low-voltage die pad **91** is electrically connected to the low-voltage circuit **20** by the wires **W1**. Thus, the low-voltage die pad **91** has the same potential as the ground **GND1** of the low-voltage circuit **20**.

(91) The high-voltage circuit chip **70** and the high-voltage leads **102** of the high-voltage lead frame **100** are electrically connected by the wires **W4**. More specifically, the second electrode pads **72** and the third electrode pads **73** of the high-voltage circuit chip **70** are connected to the high-voltage leads **102** by the wires **W4**. Thus, the high-voltage circuit **30** is electrically connected to the high-voltage leads **102** (external electrodes of the gate driver **10** that are electrically connected to the switching element **501** or the like). In the present embodiment, the two high-voltage leads **102** integrated with the high-voltage die pad **101** include a ground terminal, and the high-voltage die pad **101** is electrically connected to the high-voltage circuit **30** by the wires **W4**. Thus, the high-voltage die pad **101** has the same potential as the ground **GND2** of the high-voltage circuit **30**.

(92) The transformer chip **80** is connected to the low-voltage circuit chip **60** by the wires **W2** and is also connected to the high-voltage circuit chip **70** by the wires **W3**. More specifically, the first electrode pads **81** of the transformer chip **80** and the second electrode pads **62** of the low-voltage circuit chip **60** are connected by the wires **W2**. More specifically, the second electrode pads **82** of the transformer chip **80** and the first electrode pads **71** of the high-voltage circuit chip **70** are connected by the wires **W3**.

(93) The first coil **41A** of the transformer **40A** and the first coil **41B** of the transformer **40B** (refer to FIG. 1) are electrically connected to the ground **GND1** of the low-voltage circuit **20** through the wires **W2**, the low-voltage circuit chip **60**, and the like. The second coil **42A** of the transformer **40A** and the second coil **42B** of the transformer **40B** (refer to FIG. 1) is electrically connected to the ground **GND2** of the high-voltage circuit **30** through the wires **W3**, the high-voltage circuit chip **70**, and the like.

(94) Structure of Transformer Chip

(95) An example of the structure of the transformer chip **80** will now be described with reference to FIGS. 3 to 11.

(96) In the following description, a direction from the chip back surface **80r** of the transformer chip **80** toward the chip main surface **80s** is referred to as an upward direction, and a direction from the chip main surface **80s** toward the chip back surface **80r** is referred to as a downward direction.

(97) FIG. 3 is a perspective view showing the outer appearance of the transformer chip **80**.

(98) FIG. 4 is a plan view of the transformer chip **80**. The transformers **40A** and **40B**, the capacitors **50A** and **50B**, a shield electrode **86**, and dummy patterns **120** and **125**, which will be described later, are indicated by broken lines for the sake of illustration.

(99) FIG. 5 is a cross-sectional view of the transformer chip **80** that is cut along an xy-plane through a position of the first coils **41A** and **41B** in the z-direction showing the connection relationship of the first coils **41A** and **41B**. FIG. 6 is a cross-sectional view of the transformer chip **80** that is cut along an xy-plane through a position of the second coils **42A** and **42B** in the z-direction showing the connection relationship of the second coils **42A** and **42B**. FIGS. 5 and 6 do not show hatching for the sake of convenience.

(100) FIG. 7 is a cross-sectional view of the transformer chip **80** that is cut along an xy-plane through a position of the first capacitor electrodes **51A** and **51B** in the z-direction. FIG. 8 is a cross-sectional view of the transformer chip **80** that is cut along an xy-plane through a position of the second capacitor electrodes **52A** and **52B** in the z-direction. FIGS. 7 and 8 do not show hatching for the sake of convenience.

(101) FIG. 9 is a cross-sectional view of the transformer chip **80** taken along line 9-9 in FIG. 4 showing a cross-sectional structure of the transformer **40A** and the capacitor. FIG. 9 partially does not show hatching for simplicity and clarity. FIG. 10 is an enlarged partial view of FIG. 9 showing portions of the first coil **41A** and the first capacitor electrode **51A**. FIG. 11 is an enlarged partial view of FIG. 9 showing portions of the second coil **42A** and the second capacitor electrode **52A**.

FIGS. 10 and 11 partially do not show hatching in the same manner as FIG. 9.

(102) As shown in FIG. 4, the transformer chip 80 of the present embodiment includes two sets of the transformers 40A and 40B and the capacitors 50A and 50B. More specifically, the transformer chip 80 is a semiconductor chip that includes the transformers 40A and 40B and the capacitors 50A and 50B in the single chip. Thus, the transformer chip 80 is arranged separately from the low-voltage circuit chip 60 and the high-voltage circuit chip 70 (refer to FIG. 2).

(103) The two sets of the transformers 40A and 40B and the capacitors 50A and 50B have the same structure. Furthermore, the transformer 40B and the transformer 40A have the same structure. The capacitor 50B and the capacitor 50A have the same structure. The structures of the transformer 40A and the capacitor 50A will be described in detail. The structures of the transformer 40B and the capacitor 50B will not be described in detail.

(104) As shown in FIG. 4, the transformer chip 80 includes four chip side surfaces 80a, 80b, 80c, and 80d that are orthogonal to the chip main surface 80s and the chip back surface 80r. The chip side surfaces 80a to 80d are arranged between the chip main surface 80s and the chip back surface 80r in the z-direction. The chip side surfaces 80a and 80b define opposite surfaces of the transformer chip 80 in the y-direction. The chip side surfaces 80c and 80d define opposite surfaces of the transformer chip 80 in the x-direction. In plan view, the chip side surfaces 80a and 80b define long sides of the transformer chip 80. The chip side surfaces 80c and 80d define short sides of the transformer chip 80. In the present embodiment, the chip side surface 80a is located closer to the high-voltage circuit chip 70 (refer to FIG. 2) than the chip side surface 80b is. The chip side surface 80b is located closer to the low-voltage circuit chip 60 (refer to FIG. 2) than the chip side surface 80a is.

(105) As shown in FIGS. 4 and 9, the transformer chip 80 includes a substrate 83 and an insulation layer 84 formed on the substrate 83.

(106) The substrate 83 is composed of, for example, a semiconductor substrate. In the present embodiment, the substrate 83 is formed from a material including silicon (Si). As the semiconductor substrate, a wide-bandgap semiconductor or a compound semiconductor may be used for the substrate 83. Alternatively, instead of the semiconductor substrate, an insulating substrate formed from a glass-containing material may be used for the substrate 83.

(107) The wide-bandgap semiconductor is a semiconductor substrate having a band gap that is greater than or equal to 2.0 eV. The wide-bandgap semiconductor may be silicon carbide (SiC). The compound semiconductor may be a group III-V compound semiconductor. The compound semiconductor may include at least one of aluminum nitride (AlN), indium nitride (InN), gallium nitride (GaN), and gallium arsenide (GaAs).

(108) The substrate 83 includes a substrate main surface 83s and a substrate back surface 83r facing opposite directions in the z-direction. The substrate back surface 83r defines the chip back surface 80r of the transformer chip 80.

(109) As shown in FIG. 9, the insulation layer 84 of the present embodiment includes insulation films 85 stacked on the substrate main surface 83s of the substrate 83 in the z-direction. Thus, the z-direction may be referred to as a thickness-wise direction of the insulation layer 84. Also, the z-direction may be referred to as a thickness-wise direction of the insulation films 85. The insulation layer 84 is formed on the substrate main surface 83s of the substrate 83.

(110) Each insulation film 85 includes a first insulation film 85A and a second insulation film 85B formed on the first insulation film 85A. Thus, the insulation layer 84 includes multiple first insulation films 85A and multiple second insulation films 85B. The first insulation films 85A and the second insulation films 85B are alternately stacked in the z-direction.

(111) The first insulation film 85A is a thin film and is, for example, an etching stopper layer. The first insulation film 85A is formed from a material including silicon nitride (SiN), SiC, nitrogen-added silicon carbide (SiCN), or the like. In the present embodiment, the first insulation film 85A is formed from a material including SiN. The second insulation film 85B is, for example, an

interlayer insulation film. The second insulation film **85B** is formed from a material including silicon oxide (SiO₂). As shown in FIG. **9**, the second insulation film **85B** is greater in thickness than the first insulation film **85A**. The thickness of the first insulation film **85A** may be in a range of 100 nm to 1000 nm. The thickness of the second insulation film **85B** may be in a range of 1000 nm to 3000 nm. In the present embodiment, the thickness of the first insulation film **85A** is, for example, approximately 300 nm, and the thickness of the second insulation film **85B** is, for example, approximately 2000 nm.

(112) The second insulation films **85B** include a lowermost insulation film **85L**, which is in contact with the substrate main surface **83s** of the substrate **83**, and an uppermost insulation film **85U**.

Thus, each of the lowermost insulation film **85L** and the uppermost insulation film **85U** is smaller in thickness than other insulation films **85**. The thickness of each of the lowermost insulation film **85L** and the uppermost insulation film **85U** is in a range of the thickness of the first insulation film **85A** to the thickness of the second insulation film **85B**.

(113) The thickness of the lowermost insulation film **85L** and the uppermost insulation film **85U** may be changed in any manner. In an example, the thickness of each of the lowermost insulation film **85L** and the uppermost insulation film **85U** may be greater than the thickness of the second insulation film **85B** or may be greater than or equal to the thickness of an insulation film **85** that includes the first insulation film **85A** and the second insulation film **85B**.

(114) The transformer chip **80** includes the shield electrode **86** formed in the insulation layer **84**. The shield electrode **86** limits entrance of moisture into the insulation layer **84** and formation of cracks in the insulation layer **84**. In plan view, the shield electrode **86** is arranged on a peripheral portion of the insulation layer **84** (peripheral portion of the transformer chip **80**). More specifically, as shown in FIGS. **4** to **8**, the shield electrode **86** is spaced apart from the chip side surfaces **80a** to **80d**. In plan view, the shield electrode **86** is strip-shaped and extends along the chip side surfaces **80a** to **80d**. In the present embodiment, the shield electrode **86** has the form of a rectangular loop in plan view. The shield electrode **86** divides the insulation layer **84** into an inner region **87** and an outer region **88**. In the present embodiment, as shown in FIG. **9**, the uppermost insulation film **85U** extends across the shield electrode **86** in plan view. That is, the uppermost insulation film **85U** includes the outer region **88**.

(115) As shown in FIG. **4**, the inner region **87** is a region of the insulation layer **84** protected by the shield electrode **86**. In plan view, the inner region **87** is rectangular so that the long sides extend in the x-direction and the short sides extend in the y-direction. The outer region **88** is a rectangular loop region surrounding the inner region **87** in plan view. The outer region **88** is located between the shield electrode **86** and each of the chip side surfaces **80a** to **80d** in plan view. Thus, the outer region **88** is a rectangular loop region including the chip side surfaces **80a** to **80d**.

(116) As shown in FIG. **9**, the shield electrode **86** extends through the insulation films **85** in the z-direction. More specifically, as viewed in a direction orthogonal to the z-direction, the shield electrode **86** is arranged to overlap the coils **41A**, **41B**, **42A**, and **42B** of the transformers **40A** and **40B** and the capacitor electrodes **51A**, **51B**, **52A**, and **52B** of the capacitors **50A** and **50B**. In the present embodiment, the shield electrode **86** extends in the z-direction from the insulation film **85** that is located immediately below the uppermost insulation film **85U** to the insulation film **85** that is located immediately above the lowermost insulation film **85L**. The shield electrode **86** is formed from a material including one or more selected from titanium (Ti), titanium nitride (TiN), Au, Ag, Cu, Al, and tungsten (W).

(117) The lowermost insulation film **85L** includes vias **89** extending through the lowermost insulation film **85L** in the z-direction. The vias **89** are arranged to overlap the shield electrode **86** in plan view and connect the shield electrode **86** to the substrate **83**. Thus, the shield electrode **86** is electrically connected to the substrate **83**. In an example, the vias **89** and the shield electrode **86** may be formed from the same material.

(118) As shown in FIG. **4**, the transformers **40A** and **40B** and the capacitors **50A** and **50B** are

embedded in the insulation layer **84**. The transformers **40A** and **40B** and the capacitors **50A** and **50B** are arranged in the inner region **87**. As viewed in the z-direction, the capacitor **50A** overlaps the transformer **40A**, and the capacitor **50B** overlaps the transformer **40B**. The transformers **40A** and **40B** and the capacitors **50A** and **50B** are aligned with each other in the y-direction and spaced apart from each other in the x-direction. In other words, the transformers **40A** and **40B** and the capacitors **50A** and **50B** are arranged in a direction orthogonal to a direction in which the chips **60**, **70**, and **80** are arranged.

(119) As shown in FIG. 4, the transformer **40A** and the capacitor **50A** alternate with the transformer **40B** and the capacitor **50B** in the x-direction from the chip side surface **80c** toward the chip side surface **80d**.

(120) As shown in FIG. 5, the first coil **41A** of the transformer **40A** includes a first coil wiring **43A**, a first ground terminal **45** connected to one end of the first coil wiring **43A**, and a first signal terminal **44A** connected to the other end of the first coil wiring **43A**. The first coil **41B** of the transformer **40B** includes a first coil wiring **43B**, a first ground terminal **45** connected to one end of the first coil wiring **43B**, and a first signal terminal **44B** connected to the other end of the first coil wiring **43B**. The first ground terminal **45** is configured to be a common terminal to the first coil **41A** and the first coil **41B**. Alternatively, the first coil **41A** and the first coil **41B** may be provided with separate first ground terminals.

(121) The first coil wirings **43A** and **43B** are elliptical and spiral in plan view. The first signal terminals **44A** and **44B** are arranged at an inner side of the first coil wirings **43A** and **43B**. The first ground terminal **45** is arranged between the first coil **41A** of the transformer **40A** and the first coil **41B** of the transformer **40B**. The first coils **41A** and **41B** are formed from a material including one or more selected from Ti, TiN, Au, Ag, Cu, Al, and W.

(122) The first signal terminal **44A** is connected by an interconnect **131A** to a first electrode pad **81A**, shown in FIG. 4. The first signal terminal **44B** is connected by an interconnect **131B** to a first electrode pad **81B**, shown in FIG. 4. The first ground terminal **45** is connected by an interconnect **131C** to a first electrode pad **81C**, shown in FIG. 4.

(123) As shown in FIG. 6, the second coil **42A** of the transformer **40A** includes a second coil wiring **46A**, a second ground terminal **48** connected to one end of the second coil wiring **46A**, and a second signal terminal **47A** connected to the other of the second coil wiring **46A**. The second coil **42B** of the transformer **40B** includes a second coil wiring **46B**, a second ground terminal **48** connected to one end of the second coil wiring **46B**, and a second signal terminal **47B** connected to the other end of the second coil wiring **46B**. The second ground terminal **48** is configured to be a common terminal to the second coil **42A** and the second coil **42B**. Alternatively, the second coil **42A** and the second coil **42B** may be provided with separate second ground terminals.

(124) The second coil wirings **46A** and **46B** are elliptical and spiral in plan view. The second signal terminals **47A** and **47B** are arranged at an inner side of the second coil wirings **46A** and **46B**. The second ground terminal **48** is arranged between the second coil **42A** of the transformer **40A** and the second coil **42B** of the transformer **40B**. The second coils **42A** and **42B** are formed from a material including one or more selected from Ti, TiN, Au, Ag, Cu, Al, and W.

(125) The second signal terminal **47A** is connected to a second electrode pad **82A**, shown in FIG. 4. The second signal terminal **47B** is connected to a second electrode pad **82B**, shown in FIG. 4. The second ground terminal **48** is connected to a second electrode pad **82C**, shown in FIG. 4.

(126) In the present embodiment, the second coil wiring **46A** is formed in the same winding direction as the first coil wiring **43A**, shown in FIG. 5. The second coil wiring **46A** and the first coil wiring **43A** are wound the same number of times. In the present embodiment, the second coil wiring **46B** is formed in the same winding direction as the first coil wiring **43B**, shown in FIG. 5. The second coil wiring **46B** and the first coil wiring **43B** are wound the same number of times.

(127) As shown in FIG. 9, the first coil **41A** and the second coil **42A** of the transformer **40A** are opposed to each other in the z-direction with an insulation film **85** interposed. In the present

embodiment, the first coil **41A** and the second coil **42A** are opposed to each other in the z-direction with multiple insulation films **85** interposed.

(128) As shown in FIG. **9**, the first coil **41A** is configured to be a conductive layer embedded in one insulation film **85**. More specifically, the first coil **41A** is embedded in an insulation film **851**. The insulation film **851** includes a coil groove **141** (first coil groove) extending through the second insulation film **85B** in the z-direction. The conductive layer forming the first coil **41A** is embedded in the coil groove **141** of the second insulation film **85B** of the insulation film **851**. The insulation film **851**, in which the first coil **41A** is embedded, is covered by insulation films **85** located next to the insulation film **851** in the z-direction. Thus, the first coil **41A** is embedded in the insulation films **85**.

(129) As shown in FIG. **9**, the second coil **42A** is configured to be a conductive layer embedded in one insulation film **85**. More specifically, the second coil **42A** is embedded in an insulation film **854**. The insulation film **854** includes a coil groove **142** (second coil groove) extending through the second insulation film **85B** in the z-direction. The conductive layer forming the second coil **42A** is embedded in the coil groove **142** of the second insulation film **85B** of the insulation film **854**. The insulation film **854**, in which the second coil **42A** is embedded, is covered by insulation films **85** located next to the insulation film **854** in the z-direction. Thus, the second coil **42A** is embedded in the insulation films **85**.

(130) In the z-direction, the second coil **42A** is located farther away from the substrate **83** than the first coil **41A** is. In other words, the second coil **42A** is located above the first coil **41A**. Also, in other words, the first coil **41A** is located closer to the substrate **83** than the second coil **42A** is. In the present embodiment, the distance between the first coil **41A** and the second coil **42A** in the z-direction is greater than the distance between the first coil **41A** and the substrate main surface **83s** of the substrate **83**.

(131) In plan view, the first capacitor electrode **51A** of the capacitor **50A**, shown in FIG. **7**, overlaps the first coil **41A**, shown in FIG. **5**. The first capacitor electrode **51A** is formed from a conductive material. More preferably, the first capacitor electrode **51A** is formed from a nonmagnetic material. As the nonmagnetic material, one or more is selected from Ti, TiN, tungsten titanium (TiW), tantalum (Ta), (tantalum nitride) TaN, chromium (Cr), chromium silicide (CrSi), Au, Ag, Cu, Al, and W. Alternatively, the first capacitor electrode **51A** may be formed of a conductive material other than the materials described above. In the present embodiment, the first capacitor electrode **51A** is formed from a material including Ti.

(132) The first capacitor electrode **51A** includes a first electrode wiring **53A**, a first capacitor end **54A**, and a first capacitor ground terminal **55**. The first electrode wiring **53A** is elliptical and spiral in the same manner as the first coil wiring **43A**, shown in FIG. **5**. The first electrode wiring **53A** is identical in shape to the first coil wiring **43A** of the first coil **41A**, shown in FIG. **5**. That is, the first electrode wiring **53A** is set to be equal to the first coil wiring **43A** in ratio of wiring width to wiring spacing.

(133) The first electrode wiring **53A** includes a first slit **51As** extending from the center of the first electrode wiring **53A** toward an outer side of the first electrode wiring **53A**. The first electrode wiring **53A** is annular and is open by the first slit **51As**. The first slit **51As** limits formation of a current loop in the first electrode wiring **53A**.

(134) The first capacitor end **54A** overlaps the first signal terminal **44A** of the first coil **41A**, shown in FIG. **5**. In plan view, the first capacitor end **54A** is identical in shape to the first signal terminal **44A**. The first capacitor end **54A** is arranged at an inner side of the first electrode wiring **53A** and is connected to the first electrode wiring **53A**. The shape of the first capacitor end **54A** may be changed in any manner. Moreover, the first capacitor end **54A** may be omitted.

(135) The first capacitor ground terminal **55** overlaps the first ground terminal **45** of the first coil **41A**, shown in FIG. **5**. In plan view, the first capacitor ground terminal **55** is identical in shape to the first ground terminal **45**. The first capacitor ground terminal **55** is arranged between the first

capacitor electrode **51A** of the capacitor **50A** and the first capacitor electrode **51B** of the capacitor **50B**. The first capacitor ground terminal **55** is electrically connected to each interconnect part of the first electrode wiring **53A** by an interconnect **55A** extending toward the center of the first electrode wiring **53A**.

(136) In plan view, the first capacitor electrode **51B** of the capacitor **50B**, shown in FIG. 7, overlaps the first coil **41B**, shown in FIG. 5. The first capacitor electrode **51B** is formed from a conductive material. More preferably, the first capacitor electrode **51B** is formed from a nonmagnetic material. As the nonmagnetic material, one or more is selected from Ti, TiN, TiW, Ta, TaN, Cr, CrSi, Au, Ag, Cu, Al, and W. Alternatively, the first capacitor electrode **51B** may be formed of a conductive material other than the materials described above.

(137) The first capacitor electrode **51B** includes a first electrode wiring **53B**, a first capacitor end **54B**, and a first ground terminal **55**. Thus, the first capacitor electrodes **51A** and **51B** include the first ground terminal **55** as a common terminal in the same manner as the first coils **41A** and **41B**, shown in FIG. 5.

(138) The first electrode wiring **53B** is elliptical and spiral in the same manner as the first coil wiring **43B**, shown in FIG. 5. The first electrode wiring **53B** is identical in shape to the first coil wiring **43B** of the first coil **41B**, shown in FIG. 5. That is, the first electrode wiring **53B** is set to be equal to the first coil wiring **43B** in ratio of wiring width to wiring spacing. The first electrode wiring **53B** includes a first slit **51Bs** extending from the center of the first electrode wiring **53B** toward an outer side of the first electrode wiring **53B**. The first electrode wiring **53B** is annular and is open by the first slit **51Bs**. The first slit **51Bs** limits formation of a current loop in the first electrode wiring **53B**.

(139) The first capacitor end **54B** overlaps the first signal terminal **44B** of the first coil **41B**, shown in FIG. 5. In plan view, the first capacitor end **54B** is identical in shape to the first signal terminal **44B**. The first capacitor end **54B** is arranged at an inner side of the first electrode wiring **53B** and is connected to the first electrode wiring **53B**. The shape of the first capacitor end **54B** may be changed in any manner. Moreover, the first capacitor end **54B** may be omitted.

(140) The first capacitor ground terminal **55** is electrically connected to each interconnect part of the first electrode wiring **53B** by an interconnect **55B** extending toward the center of the first electrode wiring **53B**.

(141) In plan view, the second capacitor electrode **52A** of the capacitor **50A**, shown in FIG. 8, overlaps the second coil **42A**, shown in FIG. 6. The second capacitor electrode **52A** is formed from a conductive material. More preferably, the second capacitor electrode **52A** is formed from a nonmagnetic material. As the nonmagnetic material, one or more is selected from Ti, TiN, TiW, Ta, TaN, Cr, CrSi, Au, Ag, Cu, Al, and W. Alternatively, the second capacitor electrode **52A** may be formed of a conductive material other than the materials described above.

(142) The second capacitor electrode **52A** includes a second electrode wiring **56A**, a second capacitor end **57A**, and a second capacitor ground terminal **58**. The second electrode wiring **56A** is elliptical and spiral in the same manner as the second coil wiring **46A**, shown in FIG. 6. The second electrode wiring **56A** is identical in shape to the second coil wiring **46A** of the second coil **42A**, shown in FIG. 6. That is, the second electrode wiring **56A** is set to be equal to the second coil wiring **46A** in ratio of wiring width to wiring spacing.

(143) The second electrode wiring **56A** includes a second slit **52As** extending from the center of the second electrode wiring **56A** toward an outer side of the second electrode wiring **56A**. The second electrode wiring **56A** is annular and is open by the second slit **52As**. The second slit **52As** limits formation of a current loop in the second electrode wiring **56A**.

(144) The second capacitor end **57A** overlaps the second signal terminal **47A** of the second coil **42A**, which is shown in FIG. 6. In plan view, the second capacitor end **57A** is identical in shape to the second signal terminal **47A**. The second capacitor end **57A** is arranged at an inner side of the second electrode wiring **56A** and is connected to the second electrode wiring **56A**. The shape of the

second capacitor end **57A** may be changed in any manner. Moreover, the second capacitor end **57A** may be omitted.

(145) The second capacitor ground terminal **58** overlaps the second ground terminal **48** of the second coil **42A**, which is shown in FIG. **6**. In plan view, the second capacitor ground terminal **58** is identical in shape to the second ground terminal **48**. The second capacitor ground terminal **58** is arranged between the second capacitor electrode **52A** of the capacitor **50A** and the second capacitor electrode **52B** of the capacitor **50B**. The second capacitor ground terminal **58** is electrically connected to each interconnect part of the second electrode wiring **56A** by an interconnect **58A** extending toward the center of the second electrode wiring **56A**.

(146) In plan view, the second capacitor electrode **52B** of the capacitor **50B**, shown in FIG. **8**, overlaps the second coil **42B**, shown in FIG. **6**. The second capacitor electrode **52B** is formed from a conductive material. More preferably, the second capacitor electrode **52B** is formed from a nonmagnetic material. As the nonmagnetic material, one or more is selected from Ti, TiN, TiW, Ta, TaN, Cr, CrSi, Au, Ag, Cu, Al, and W. Alternatively, the second capacitor electrode **52B** may be formed of a conductive material other than the materials described above.

(147) The second capacitor electrode **52B** includes a second electrode wiring **56B**, a second capacitor end **57B**, and a second capacitor ground terminal **58**. Thus, the second capacitor electrodes **52A** and **52B** include the second capacitor ground terminal **58** as a common terminal in the same manner as the second coils **42A** and **42B**, shown in FIG. **6**.

(148) The second electrode wiring **56B** is elliptical and spiral in the same manner as the second coil wiring **46B**, shown in FIG. **6**. The second electrode wiring **56B** is identical in shape to the second coil wiring **46B** of the second coil **42B**, shown in FIG. **6**. That is, the second electrode wiring **56B** is set to be equal to the second coil wiring **46B** in ratio of wiring width to wiring spacing. The second electrode wiring **56B** includes a second slit **52Bs** extending from the center of the second electrode wiring **56B** toward an outer side of the second electrode wiring **56B**. The second electrode wiring **56B** is annular and is open by the second slit **52Bs**. The second slit **52Bs** limits formation of a current loop in the second electrode wiring **56B**.

(149) The second capacitor end **57B** overlaps the second signal terminal **47B** of the second coil **42B**, shown in FIG. **6**. In plan view, the second capacitor end **57B** is identical in shape to the second signal terminal **47B**. The second capacitor end **57B** is arranged at an inner side of the second electrode wiring **56B** and is connected to the second electrode wiring **56B**. The shape of the second capacitor end **57B** may be changed in any manner. Moreover, the second capacitor end **57B** may be omitted.

(150) The second capacitor ground terminal **58** is electrically connected to each interconnect part of the second electrode wiring **56B** by an interconnect **58B** extending toward the center of the second electrode wiring **56B**.

(151) As shown in FIG. **9**, the first capacitor electrode **51A** and the second capacitor electrode **52A** of the capacitor **50A** are arranged between the first coil **41A** and the second coil **42A** of the transformer **40A**. The first capacitor electrode **51A** and the second capacitor electrode **52A** are opposed to each other in the z-direction with multiple insulation films **85** interposed.

(152) In the present embodiment, the first capacitor electrode **51A** is formed in an insulation film **852** located next to the insulation film **851**, in which the first coil **41A** is formed. In the present embodiment, the second capacitor electrode **52A** is formed in an insulation film **853** located next to the insulation film **854**, in which the second coil **42A** is formed.

(153) The first capacitor electrode **51A** is configured to be a conductive layer embedded in one insulation film **85**. More specifically, the insulation film **852**, in which the first capacitor electrode **51A** is embedded, includes a first insulation film **85A** and a second insulation film **85B**. The conductive layer forming the first capacitor electrode **51A** is arranged between the first insulation film **85A** and the second insulation film **85B** of the insulation film **852**. More specifically, one surface (lower surface) of the first insulation film **85A** in the z-direction is in contact with the first

coil **41A**, and the other surface (upper surface) is in contact with the first capacitor electrode **51A**. The first insulation film **85A** of the insulation film **852** is arranged between the first capacitor electrode **51A** and the first coil **41A**. The first insulation film **85A** of the insulation film **852** corresponds to a first thin film. The second insulation film **85B** of the insulation film **852** corresponds to a first interlayer insulation film. Thus, the first capacitor electrode **51A** is embedded in the insulation film **85**.

(154) The second capacitor electrode **52A** is configured to be a conductive layer embedded in one insulation film **85**. More specifically, the insulation film **853**, in which the second capacitor electrode **52A** is embedded, includes a first insulation film **85A** and a second insulation film **85B**. The insulation film **854**, in which the second coil **42A** is embedded, includes a first insulation film **85A** and a second insulation film **85B**. The first insulation film **85A** of the insulation film **854** is in contact with the second insulation film **85B** of the insulation film **853**. The conductive layer forming the second capacitor electrode **52A** is arranged between the second insulation film **85B** of the insulation film **853** and the first insulation film **85A** of the insulation film **854** located next to the insulation film **853**. More specifically, one surface (lower surface) of the first insulation film **85A** in the z-direction is in contact with the second capacitor electrode **52A**, and the other surface (upper surface) is in contact with the second coil **42A**. The first insulation film **85A** of the insulation film **854** is arranged between the second capacitor electrode **52A** and the second coil **42A**. The first insulation film **85A** of the insulation film **854** corresponds to a second thin film. The second insulation film **85B** of the insulation film **853** corresponds to a second interlayer insulation film. Thus, the second capacitor electrode **52A** is embedded in the insulation films **85**.

(155) The distance between the first capacitor electrode **51A** and the second capacitor electrode **52A** is determined by the thickness of the insulation film **85** and the number of the insulation films **85** stacked between the first capacitor electrode **51A** and the second capacitor electrode **52A**. The distance is set in accordance with the insulation voltage or electric field strength of the transformer chip **80**.

(156) As shown in FIG. **9**, the first capacitor electrode **51A** is electrically connected to the first coil **41A**.

(157) The first coil **41A** includes the first coil wiring **43A**, the first signal terminal **44A**, and the first ground terminal **45**. The first capacitor electrode **51A** includes the first electrode wiring **53A**, the first capacitor end **54A**, and the first capacitor ground terminal **55**. The first electrode wiring **53A** overlaps the first coil wiring **43A** in the z-direction. The first capacitor end **54A** overlaps the first signal terminal **44A** in the z-direction. The first capacitor ground terminal **55** overlaps the first ground terminal **45** in the z-direction.

(158) The first capacitor ground terminal **55** of the first capacitor electrode **51A** is connected to the first ground terminal **45** of the first coil **41A**. The first insulation film **85A** of the insulation film **852**, in which the first capacitor electrode **51A** is embedded, includes a first opening **145** exposing the first ground terminal **45** of the first coil **41A**. The first capacitor ground terminal **55** of the first capacitor electrode **51A** is in contact with the first ground terminal **45** of the first coil **41A** through the first opening **145**. Thus, the first capacitor ground terminal **55** of the first capacitor electrode **51A** is electrically connected to the first ground terminal **45** of the first coil **41A** through the first opening **145**.

(159) As shown in FIG. **9**, the second capacitor electrode **52A** is electrically connected to the second coil **42A**.

(160) The second coil **42A** includes the second coil wiring **46A**, the second signal terminal **47A**, and the second ground terminal **48**. The second capacitor electrode **52A** includes the second electrode wiring **56A**, the second capacitor end **57A**, and the second capacitor ground terminal **58**. The second electrode wiring **56A** overlaps the second coil wiring **46A** in the z-direction. The second capacitor end **57A** overlaps the second signal terminal **47A** in the z-direction. The second capacitor ground terminal **58** overlaps the second ground terminal **48** in the z-direction.

(161) The second capacitor ground terminal **58** of the second capacitor electrode **52A** is connected to the second ground terminal **48** of the second coil **42A**. The first insulation film **85A** of the insulation film **854**, in which the second coil **42A** is embedded, includes a second opening **146** partially exposing the second capacitor ground terminal **58** of the second capacitor electrode **52A**. The second ground terminal **48** of the second coil **42A** is in contact with the second capacitor ground terminal **58** of the second capacitor electrode **52A** through the second opening **146**. Thus, the second ground terminal **48** of the second coil **42A** is electrically connected to the second capacitor ground terminal **58** of the second capacitor electrode **52A** through the second opening **146**.

(162) As shown in FIG. **4**, in plan view, the first electrode pads **81** and the second electrode pads **82** are arranged in the inner region **87**.

(163) As shown in FIG. **9**, the electrode pads **81** and **82** are formed on the uppermost insulation film **85U**. In other words, the electrode pads **81** and the electrode pads **82** are embedded in the insulation film **85**. In the present embodiment, the electrode pad **81** and the electrode pad **82** are located farther away from the substrate **83** than the second coils **42A** and **42B** of the transformers **40A** and **40B**. In other words, the electrode pad **81** and the electrode pad **82** are located above the second coils **42A** and **42B** of the transformers **40A** and **40B**. In the present embodiment, the distance between the first coil **41A** and the second coil **42A** is greater than the distance between the second coil **42A** and each of the electrode pads **81** and **82** in the z-direction.

(164) As shown in FIG. **4**, in plan view, each first electrode pad **81** is aligned with one of the two transformers **40A** or one of the two transformers **40B** in the x-direction or arranged between the transformer **40A** and the transformer **40B** in the x-direction. The first electrode pads **81** are arranged closer to the chip side surface **80b** than the transformers **40A** and **40B** are in the y-direction. In other words, the first electrode pads **81** are arranged between the chip side surface **80b** and the transformers **40A** and **40B** in the y-direction. In other words, in plan view, the first electrode pads **81** are arranged closer to the low-voltage leads **92** (refer to FIG. **2**) than the transformers **40A** and **40B** are.

(165) In the description hereafter, the first electrode pad **81** that is aligned with the transformer **40A** in the x-direction is referred to as a first electrode pad **81A** for the sake of brevity. The first electrode pad **81** that is aligned with the transformer **40B** in the x-direction is referred to as a first electrode pad **81B** for the sake of brevity. The first electrode pad **81** arranged between the transformer **40A** and the transformer **40B** in the x-direction is referred to as a first electrode pad **81C**. The first electrode pads **81A** to **81C** will be referred to as the first electrode pads **81** when the description is common to the first electrode pads **81A** to **81C**.

(166) The first electrode pads **81A** are arranged to overlap the transformer **40A** as viewed in the y-direction. The first electrode pads **81B** are arranged to overlap the transformer **40B** as viewed in the y-direction. The first electrode pads **81C** are arranged to overlap a portion located between the transformer **40A** and the transformer **40B** in the x-direction as viewed in the y-direction. The first electrode pads **81A** to **81C** are aligned with each other in the y-direction and spaced apart from each other in the x-direction.

(167) As shown in FIG. **4**, in plan view, the first electrode pads **82** are arranged in the transformers **40A** and **40B** and between the transformer **40A** and the transformer **40B** in the x-direction. As viewed in the x-direction, the second electrode pads **82** are arranged to overlap the transformers **40A** and **40B**. In the description hereafter, the second electrode pad **82** arranged in the transformer **40A** is referred to as a second electrode pad **82A**, the second electrode pad **82** arranged in the transformer **40B** is referred to as a second electrode pad **82B**, and the second electrode pad arranged between the transformer **40A** and the transformer **40B** is referred to as the second electrode pad **82C**. The second electrode pads **82A** to **82C** will be referred to as the second electrode pads **82** when the description is common to the second electrode pads **82A** to **82C**.

(168) The second electrode pads **82A** are arranged in an inner space of the transformer **40A** defined

by the elliptical, spiral second coil **42A**. The second electrode pads **82B** are arranged in an inner space of the transformer **40B** defined by the elliptical, spiral second coil **42A**. The second electrode pads **82C** are arranged between the transformer **40A** and the transformer **40B** in the x-direction. Each of the second electrode pads **82A** to **82C** is formed of two electrode pads adjacent to each other in the x-direction. The second electrode pads **82A** to **82C** are aligned with each other in the y-direction and spaced apart from each other in the x-direction.

(169) As shown in FIGS. **4**, **5**, and **9**, the first electrode pads **81A** are electrically connected to the first coil **41A** of the transformer **40A**. The first electrode pads **81C** are electrically connected to the first coil **41A** of the transformer **40A**. As shown in FIGS. **4** and **5**, the first electrode pads **81B** are electrically connected to the first coil **41B** of the transformer **40B**. The first electrode pads **81C** are electrically connected to the first coil **41B** of the transformer **40B**. That is, the first electrode pads **81C** are common to the transformer **40A** and the transformer **40B**.

(170) As shown in FIGS. **4**, **6**, and **9**, each second electrode pad **82A** is separately electrically connected to the second coil **42A** of the transformer **40A**. The second electrode pads **82C** are electrically connected to the second coil **42A** of the transformer **40A**. The second electrode pads **82B**, shown in FIGS. **4** and **6**, are electrically connected to the second coil **42B** of the transformer **40B**. The second electrode pads **82C** are electrically connected to the second coil **42B** of the transformer **40B**. That is, the second electrode pads **82C** are common to the transformer **40A** and the transformer **40B**.

(171) As shown in FIGS. **5** and **6**, the transformer chip **80** includes interconnects separately connecting the electrode pads **81A** to **81C** and **82A** to **82C** to the coils **41A**, **41B**, **42A**, and **42B** of the transformers **40A** and **40B**. In the present embodiment, the interconnects include the interconnects **131A**, **131B**, and **131C**, which connect the first electrode pads **81A** to **81C** to the first coils **41A** and **41B**. The interconnects **131A** to **131C** are arranged in the inner region **87**. The interconnects **131A** to **131C** are formed from a material including one or more selected from Ti, TiN, Au, Ag, Cu, Al, and W.

(172) As shown in FIG. **5**, the interconnect **131A** connects the first electrode pad **81A** to the first signal terminal **44A** of the first coil **41A** of the transformer **40A**. The interconnect **131B** connects the first electrode pad **81B** to the first signal terminal **44B** of the first coil **41B** of the transformer **40B**. The interconnect **131C** connects the first electrode pad **81C** to the second end of the first coil **41A** of the transformer **40A** and the second end of the first coil **41B** of the transformer **40B**. The interconnect **131C** corresponds to a first voltage interconnect. The first electrode pad **81C** corresponds to a first ground electrode. The interconnects **131A** to **131C** have the same structure. Thus, in the description hereafter, the structure of the interconnect **131A** will be described. The interconnects **131B** and **131C** will not be described in detail.

(173) As shown in FIG. **9**, the interconnect **131A** includes a first interconnect part **132A** extending through the insulation films **85** in the z-direction and a second interconnect part **133A** extending in the y-direction.

(174) The first interconnect part **132A** is arranged to overlap the first electrode pad **81A** in plan view and is connected to the first electrode pad **81A**. The first interconnect part **132A** extends through the insulation films **85** from the insulation film **85** that is located immediately below the uppermost insulation film **85U** to the insulation film **85** that is located above the lowermost insulation film **85L** with one insulation film **85** interposed. The first interconnect part **132A** includes flat interconnect pieces and vias. The interconnect pieces are located at the same position as the insulation films **851** and **854**, in which the coils **41A** and **42A** are arranged. The vias are arranged between the two interconnect pieces in the z-direction, between the upper interconnect piece and the first electrode pad **81A**, and between the lower interconnect piece and the second interconnect part **133A**.

(175) The second interconnect part **133A** is arranged closer to the substrate **83** than the first interconnect part **132A** is. The second interconnect part **133A** is arranged closer to the substrate **83**

than the first coil **41A** is. In the present embodiment, the second interconnect part **133A** is arranged in the insulation film **85** located immediately above the lowermost insulation film **85L** among the insulation films **85**. Of opposite ends of the second interconnect part **133A** in the x-direction, a first end is located closer to the chip side surface **80b** of the transformer chip **80** and overlaps the first interconnect part **132A** in plan view. The second interconnect part **133A** is connected to the first interconnect part **132A**. The second interconnect part **133A** includes a second end opposite to the first end. In plan view, the second end is arranged to overlap the first coil **41A** of the transformer **40A**. More specifically, in plan view, the second end overlaps the first signal terminal **44A** of the first coil **41A** of the transformer **40A**. The second interconnect part **133A** includes vias **134A** connecting the second interconnect part **133A** and the first signal terminal **44A**.

(176) As shown in FIG. 9, the interconnect **131C** includes a first interconnect part **132C** extending through the insulation films **85** in the z-direction and a second interconnect part **133C** extending in the y-direction.

(177) The first interconnect part **132C** and the first interconnect part **132A** of the interconnect **131A** have the same structure.

(178) The first interconnect part **132C** is arranged to overlap the first electrode pad **81C** in plan view and is connected to the first electrode pad **81C**. The first interconnect part **132C** extends through the insulation films **85** from the insulation film **85** that is located immediately below the uppermost insulation film **85U** to the insulation film **85** that is located above the lowermost insulation film **85L** with one insulation film **85** interposed. The first interconnect part **132C** includes flat interconnect pieces and vias. The interconnect pieces are located at the same position as the insulation films **851** and **854**, in which the coils **41A** and **42A** are arranged. The vias are arranged between the two interconnect pieces in the z-direction, between the upper interconnect piece and the first electrode pad **81A**, and between the lower interconnect piece and the second interconnect part **133C**.

(179) The second interconnect part **133C** is arranged closer to the substrate **83** than the first interconnect part **132C** is. The second interconnect part **133C** is arranged closer to the substrate **83** than the first coil **41A** is. In the present embodiment, the second interconnect part **133C** is arranged in the insulation film **85** located immediately above the lowermost insulation film **85L** among the insulation films **85**. Of opposite ends of the second interconnect part **133C** in the x-direction, a first end is located closer to the chip side surface **80b** of the transformer chip **80** and overlaps the first interconnect part **132C** in plan view. The second interconnect part **133C** is connected to the first interconnect part **132C**. The second interconnect part **133C** includes a second end opposite to the first end. In plan view, the second end is arranged to overlap the first coil **41A** of the transformer **40A**. More specifically, in plan view, the second end overlaps the first ground terminal **45** of the first coil **41A** of the transformer **40A**. The second interconnect part **133C** includes vias **134C** connecting the second interconnect part **133C** and the first signal terminal **44A**. The second interconnect part **133C** of the interconnect **131C** is electrically connected to the substrate **83** by vias **136** that extend through the lowermost insulation film **85L**. The vias **136** may be omitted.

(180) As shown in FIG. 9, the second electrode pad **82A** is electrically connected to the second signal terminal **47A** of the second coil **42A** by vias **135A** extending through the uppermost insulation film **85U**.

(181) The second electrode pad **82C** is electrically connected to the second ground terminal **48** of the second coil **42A** by vias **135C** extending through the uppermost insulation film **85U**.

(182) As shown in FIGS. 4 and 6, in the present embodiment, the transformer chip **80** includes a dummy pattern **120** arranged around the second coils **42A** and **42B** of the transformers **40A** and **40B**. The dummy pattern **120** is a dummy coil pattern.

(183) The dummy pattern **120** is arranged in the inner region **87** and includes a first dummy pattern **121**, a second dummy pattern **122**, and a third dummy pattern **123**. The dummy patterns **121** to **123** are formed from a material including one or more selected from Ti, TiN, Au, Ag, Cu, Al, and W.

(184) In plan view, the first dummy pattern **121** is arranged in a region between the second coil **42A** of the transformer **40A** and the second coil **42B** of the transformer **40B** in the x-direction. The first dummy pattern **121** is patterned differently from the second coils **42A** and **42B**. The first dummy pattern **121** is electrically connected to the second ground terminal **48** of the second coil **42A**. The first dummy pattern **121** may be electrically connected to the second ground terminal **48** of at least one of the two second coils **42A**. Thus, the first dummy pattern **121** has the same potential as the second coils **42A** and **42B**. Therefore, as the second reference potential of the second coils **42A** and **42B** changes, the voltage of the first dummy pattern **121** may become higher than that of the first coil **41B** in the same manner as the second coil **42B**.

(185) Although not shown, the first dummy pattern **121** is aligned with the second coils **42A** and **42B** in the z-direction. Thus, the first dummy pattern **121** is located farther away from the substrate **83** than the first coils **41A** and **41B** are. In other words, the dummy pattern **120** is arranged around coils of the transformers **40A** and **40B** located closer to the chip main surface **80s** of the transformer chip **80**.

(186) The first dummy pattern **121** and the second coils **42A** and **42B** have the same voltage. This limits voltage drops between the first dummy pattern **121** and the second coils **42A** and **42B**. Thus, concentration of an electric field on the second coils **42A** and **42B** is limited.

(187) As shown in FIG. 6, the third dummy pattern **123** is formed to surround the second coils **42A** and **42B** of the transformers **40A** and **40B** in plan view. The third dummy pattern **123** is electrically connected to the first dummy pattern **121**. Thus, in the third dummy pattern **123**, similar to the first dummy pattern **121**, as the second reference potential of the second coil **42B** changes, the voltage of the third dummy pattern **123** may become higher than that of the first coil **41B**.

(188) As shown in FIG. 9, the third dummy pattern **123** is aligned with the second coil **42A** in the z-direction. Although not shown, the third dummy pattern **123** is aligned with the second coil **42B** in the z-direction. Thus, the third dummy pattern **123** is located farther away from the substrate **83** than the first coils **41A** and **41B** are. As described above, the dummy patterns **121** to **123** are aligned with each other in the z-direction.

(189) The third dummy pattern **123** and the second coils **42A** and **42B** have the same voltage. This limits voltage drops between the third dummy pattern **123** and the second coils **42A** and **42B**. Thus, concentration of an electric field on the second coils **42A** and **42B** is limited.

(190) As shown in FIG. 6, the second dummy pattern **122** is formed to surround the third dummy pattern **123** in plan view. The second dummy pattern **122** is independent of the second coils **42A** and **42B**. In other words, the second dummy pattern **122** is not electrically connected to the second coils **42A** and **42B**.

(191) As shown in FIG. 9, the second dummy pattern **122** is aligned with the second coil **42A** in the z-direction. Although not shown, the second dummy pattern **122** is aligned with the second coil **42B** in the z-direction. Thus, the second dummy pattern **122** is located farther away from the substrate **83** than the first coils **41A** and **41B** are. The second dummy pattern **122** limits an increase in the electric field strength around the second coils **42A** and **42B** and limits concentration of the electric field on the second electrode pads **82A** to **82C**.

(192) As shown in FIG. 8, in the present embodiment, the transformer chip **80** includes a dummy pattern **125** arranged around the second capacitor electrodes **52A** and **52B** of the capacitors **50A** and **50B**. The dummy pattern **125** and the dummy pattern **120**, described above, have the same structure. The dummy pattern **125** includes a slit extending from the inside of the dummy pattern **125** toward the outside of the dummy pattern **125**. The slit limits formation of a current loop in the dummy pattern **125**. The dummy pattern **125** is a dummy electrode pattern.

(193) More specifically, the dummy pattern **125** is arranged in the inner region **87** and includes a first dummy pattern **126**, a second dummy pattern **127**, and a third dummy pattern **128**. In an example, the dummy patterns **126** to **128** and the second capacitor electrode **52A** are formed from the same material.

(194) In plan view, the first dummy pattern **126** is arranged in a region between the second capacitor electrode **52A** of the capacitor **50A** and the second capacitor electrode **52B** of the capacitor **50B** in the x-direction. The first dummy pattern **126** is patterned differently from the second capacitor electrodes **52A** and **52B**. The first dummy pattern **126** is electrically connected to the second capacitor ground terminal **58** of the second capacitor electrode **52A**. The first dummy pattern **126** may be electrically connected to the second capacitor ground terminal **58** of at least one of the two second capacitor electrodes **52A**. Thus, the first dummy pattern **126** has the same potential as the second capacitor electrodes **52A** and **52B**. Therefore, as the second reference potential of the second capacitor electrodes **52A** and **52B** changes, the voltage of the first dummy pattern **126** may become higher than that of the first capacitor electrode **51B** in the same manner as the second capacitor electrode **52B**.

(195) Although not shown, the first dummy pattern **126** is aligned with the second capacitor electrodes **52A** and **52B** in the z-direction. Thus, the first dummy pattern **126** is located farther away from the substrate **83** than the first capacitor electrodes **51A** and **51B** are. In other words, the dummy pattern **125** is arranged around coils of the capacitors **50A** and **50B** located closer to the chip main surface **80s** of the transformer chip **80**.

(196) When the first dummy pattern **126** and the second capacitor electrodes **52A** and **52B** have the same voltage, a voltage drop between the first dummy pattern **126** and the second capacitor electrodes **52A** and **52B** is limited. This limits concentration of an electric field on the second capacitor electrodes **52A** and **52B**.

(197) As shown in FIG. **8**, the third dummy pattern **128** is formed to surround the second capacitor electrodes **52A** and **52B** of the capacitors **50A** and **50B** in plan view. The third dummy pattern **128** is electrically connected to the first dummy pattern **126**. Thus, in the third dummy pattern **128**, similar to the first dummy pattern **126**, as the second reference potential of the second capacitor electrode **52B** changes, the voltage of the third dummy pattern **128** may become higher than the voltage of the first capacitor electrode **51B**.

(198) As shown in FIG. **9**, the third dummy pattern **128** is aligned with the second capacitor electrode **52A** in the z-direction. Although not shown, the third dummy pattern **128** is also aligned with the second capacitor electrode **52B** in the z-direction. Thus, the third dummy pattern **128** is located farther away from the substrate **83** than the first capacitor electrodes **51A** and **51B** are. As described above, the dummy patterns **126** to **128** are aligned with each other in the z-direction.

(199) When the third dummy pattern **128** and the second capacitor electrodes **52A** and **52B** have the same voltage, a voltage drop between the third dummy pattern **128** and the second capacitor electrodes **52A** and **52B** is limited. This limits concentration of an electric field on the second capacitor electrodes **52A** and **52B**.

(200) As shown in FIG. **8**, the second dummy pattern **127** is formed to surround the third dummy pattern **128** in plan view. The second dummy pattern **127** is independent of the second capacitor electrodes **52A** and **52B**. In other words, the second dummy pattern **127** is not electrically connected to the second capacitor electrodes **52A** and **52B**.

(201) As shown in FIG. **9**, the second dummy pattern **127** is aligned with the second capacitor electrode **52A** in the z-direction. Although not shown, the second dummy pattern **127** is also aligned with the second capacitor electrode **52B** in the z-direction. Thus, the second dummy pattern **127** is located farther away from the substrate **83** than the first capacitor electrodes **51A** and **51B** are. The second dummy pattern **127** limits an increase in the electric field strength around the second capacitor electrodes **52A** and **52B** and limits concentration of the electric field on the second electrode pads **82A** to **82C**.

(202) As shown in FIG. **9**, the transformer chip **80** includes a protection film **150** and a passivation film **160**. The protection film **150** is formed on a surface **84s** of the insulation layer **84**. The protection film **150** protects the insulation layer **84**. The protection film **150** is a film formed from a material including, for example, silicon oxide.

(203) The passivation film **160** is a surface protection film for the transformer chip **80**. The passivation film **160** is formed from a material including, for example, silicon nitride. The material including silicon nitride includes, for example, SiN and SiCN. In the present embodiment, the passivation film **160** is formed from a material including SiN. The passivation film **160** defines the chip main surface **80s** of the transformer chip **80**.

(204) The first electrode pads **81** and the second electrode pads **82** are covered by the protection film **150** and the passivation film **160**. The protection film **150** and the passivation film **160** each include openings partially exposing the first electrode pads **81** and the second electrode pads **82**. Thus, each first electrode pad **81** includes an exposed surface configured to be connected to the wire W2. Also, each second electrode pad **82** includes an exposed surface configured to be connected to the wire W3.

(205) As shown in FIGS. **3** and **4**, the transformer chip **80** includes a resin layer **180** formed on the passivation film **160**. The resin layer **180** is formed from a material including, for example, polyimide (PI). The resin layer **180** is separated by a separation groove **183** into an inner resin layer **181** and an outer resin layer **182**. As shown in FIG. **4**, in plan view, the separation groove **183** surrounds the transformers **40A** and **40B**. The resin layer **180** includes a first resin opening **184** exposing the first electrode pad **81** and a second resin opening **185** exposing the second electrode pad **82**.

(206) FIG. **10** is an enlarged partial view of FIG. **9** showing an example of the structure of the first coil **41A** and the first capacitor electrode **51A**. In the example of the structure, the first coil **41A** is formed from Cu.

(207) As shown in FIG. **10**, the coil groove **141** is elliptical and spiral and is formed in the insulation film **851**. The coil groove **141** extends through the second insulation film **85B**. Therefore, the upper end and the lower end of the coil groove **141** are respectively open to the first insulation film **85A** of the upper insulation film **851** and the first insulation film **85A** of the insulation film **852**, in which the first coil **41A** is embedded.

(208) A barrier film **143** is formed on the wall surfaces (side wall surface and bottom wall surface) of the coil groove **141**. The barrier film **143** is formed on the side wall surface and the bottom wall surface so that the coil groove **141** has a cavity that is open upward. In the embodiment, the barrier film **143** is formed from a material including Ta, TaN, Ti, TiN, or the like. A body layer **144** is embedded in the coil groove **141** at an inner side of the barrier film **143**. This forms the second coil **42A**, which is an example of an embedded coil formed of the barrier film **143** and the body layer **144**. The body layer **144** is formed from a material including one or more selected from Cu, Al, and W.

(209) The first coil **41A** includes an upper surface that is flush with the upper surface of the insulation film **851**. As a result, the side surface, the upper surface, and the lower surface of the first coil **41A** are in contact with different insulation films **85**. More specifically, in the insulation film **851**, in which the first coil **41A** is embedded, the first insulation film **85A** and the second insulation film **85B** are in contact with the side surface of the second coil **42A**. In the insulation film **852** formed above the insulation film **851**, only the first insulation film **85A**, which is the lower layer, is in contact with the upper surface of the second coil **42A**. In the insulation film **85** located below, only the second insulation film **85B**, which is the upper layer, is in contact with the lower surface of the second coil **42A**.

(210) The first capacitor electrode **51A** is in contact with the upper surface of the first insulation film **85A** forming the insulation film **852**. The upper surface and the side surface of the first capacitor electrode **51A** are in contact with the second insulation film **85B** of the insulation film **852**. One surface (lower surface) of the first insulation film **85A** of the insulation film **852** in the z-direction is in contact with the second insulation film **85B** (interlayer insulation film). The other surface (upper surface) of the first insulation film **85A** of the insulation film **852** in the z-direction is in contact with the first capacitor electrode **51A**. The first insulation film **85A** of the insulation

film **852** corresponds to a first thin film. The second insulation film **85B** of the insulation film **852** corresponds to a first interlayer insulation film.

(211) The first insulation film **85A** of the insulation film **852** covers the second insulation film **85B** of the insulation film **851**, which is located below the insulation film **852**, and the first coil **41A**, which is embedded in the insulation film **851**. The first insulation film **85A** includes the first opening **145**, which partially exposes the upper surface of the first ground terminal **45** of the first coil **41A**. Then, a film of a conductive material is formed on the first insulation film **85A** to form the first capacitor electrode **51A**. The conductive material film is partially removed to form the first capacitor electrode **51A**. The second insulation film **85B** is formed to cover the first capacitor electrode **51A** and the first insulation film **85A**. This forms the first capacitor electrode **51A** embedded in the insulation film **852**.

(212) FIG. **11** is an enlarged partial view of FIG. **9** showing an example of the structure of the second coil **42A** and the second capacitor electrode **52A**. In the example of the structure, the second coil **42A** is formed from Cu.

(213) As shown in FIG. **11**, the coil groove **142** is elliptical and spiral and formed in the second insulation film **85B** of the insulation film **854**. The coil groove **142** extends through the second insulation film **85B**. Therefore, the upper end and the lower end of the coil groove **142** are respectively open to the first insulation film **85A** of the upper insulation film **85** and the first insulation film **85A** of the insulation film **854**, in which the second coil **42A** is embedded.

(214) The barrier film **143** is formed on the wall surface (side wall surface and bottom wall surface) of the coil groove **142**. The barrier film **143** is formed on the side wall surface and the bottom wall surface so that the coil groove **142** has a cavity that is open upward. In the embodiment, the barrier film **143** is formed from a material including Ta, TaN, Ti, TiN, or the like. A body layer **144** is embedded in the coil groove **142** at an inner side of the barrier film **143**. This forms the second coil **42A**, which is an example of an embedded coil formed of is formed of the barrier film **143** and the body layer **144**. The body layer **144** is formed from a material including one or more selected from Cu, Al, and W.

(215) The second coil **42A** includes an upper surface that is flush with the upper surface of the insulation film **854**. As a result, the side surface, the upper surface, and the lower surface of the second coil **42A** are in contact with different insulation films. More specifically, in the insulation film **85** in which the second coil **42A** is embedded, the second insulation film **85B** is in contact with the side surface of the second coil **42A**. In the insulation film **85** formed above the insulation film **85** in which the second coil **42A** is embedded, only the lower first insulation film **85A** is in contact with the upper surface of the second coil **42A**. In the insulation film **854**, in which the second coil **42A** is embedded, the first insulation film **85A** is in contact with the lower surface of the second coil **42A**.

(216) The second capacitor electrode **52A** is in contact with the lower surface of the first insulation film **85A** of the insulation film **854**, in which the second coil **42A** is embedded. The lower surface and the side surface of the second capacitor electrode **52A** are in contact with the second insulation film **85B** of the insulation film **853**. One surface (lower surface) of the first insulation film **85A** of the insulation film **853** in the z-direction is in contact with the second capacitor electrode **52A**. The other surface (upper surface) is in contact with the second insulation film **85B** (interlayer insulation film). The first insulation film **85A** of the insulation film **854** corresponds to a second thin film. The second insulation film **85B** of the insulation film **853** corresponds to a second interlayer insulation film.

(217) As shown in FIG. **11** with broken lines, the second insulation film **85B** of the insulation film **853** includes two layers of insulation film **85B1** and **85B2**. First, the insulation film **85B1** is formed on the upper surface of the first insulation film **85A** of the insulation film **853**. Then, a film of a conductive material is formed on the upper surface of the insulation films **85B1** to form the second capacitor electrode **52A**. The conductive material film is partially removed to form the second

capacitor electrode **52A**. The second capacitor electrode **52A** is further embedded in the insulation film **85B2** to form the second insulation film **85B**. The first insulation film **85A** of the insulation film **854** is formed to cover the second insulation film **85B** and the second capacitor electrode **52A**. This forms the second capacitor electrode **52A** embedded in the insulation film **85**.

(218) Although not described and shown, the dummy patterns **120** and **125**, shown in FIG. **9**, are formed in the same manner as the second coil **42A** and the second capacitor electrode **52A**. The interconnects **131A** and **131C**, shown in FIG. **9**, are formed in the same manner as the first coil **41A** and the second coil **42A** by embedding a barrier film and a body layer in a groove extending through the insulation film **85**.

(219) Operation

(220) The operation of the gate driver **10** according to the present embodiment will now be described.

Comparative Example

(221) A comparative example of the gate driver **10** of the present embodiment will now be described.

(222) FIG. **12** shows a comparative example of a gate driver **10R**. The gate driver **10R** of the comparative example does not include the capacitors **50** (capacitors **50A** and **50B**), which are shown in FIG. **1**. In the gate driver **10R**, a current $i_{\text{sub.C1}}$ flowing to parasitic capacitance **C1** between the first coil **41A** and the second coil **42A** of the transformer **40A** may produce noise on a set signal transmitted from the low-voltage circuit **20** to the high-voltage circuit **30**. This may result in an erroneous operation. Also, a current $i_{\text{sub.c2}}$ flowing to parasitic capacitance **C2** between the first coil **41B** and the second coil **42B** of the transformer **40B** may produce noise on a reset signal transmitted from the low-voltage circuit **20** to the high-voltage circuit **30**. This may result in an erroneous operation.

(223) To mitigate an erroneous operation caused by the currents $i_{\text{sub.C1}}$ and $i_{\text{sub.C2}}$, the high-voltage circuit **30** includes a circuit that masks noise. In an example, after receiving a reset signal (RESET), the mask circuit masks reception of a signal for a fixed period. This mitigates an erroneous operation caused by the currents $i_{\text{sub.C1}}$ and $i_{\text{sub.C2}}$ flowing to the second coils **42A** and **42B** due to the parasitic capacitances **C1** and **C2**.

(224) The capacitance values of the parasitic capacitances **C1** and **C2** may vary depending on each gate driver **10R** and operational state. That is, the capacitance values of the parasitic capacitances **C1** and **C2** are inconstant. Therefore, the position of noise superimposed on the set signal and the reset signal may change. In this regard, the high-voltage circuit **30** needs to set a long mask period in accordance with positions where noise may be produced. Since a signal cannot be transmitted during the mask period, fast transmission of a signal from the low-voltage circuit **20** to the high-voltage circuit **30** is hindered. The same issue arises when a signal is transmitted from the high-voltage circuit **30** to the low-voltage circuit **20**. Hence, the low-voltage circuit **20** also needs a mask circuit in the same manner as the high-voltage circuit **30**.

(225) FIG. **13** shows an operation of the gate driver **10** of the present embodiment.

(226) As described above, the gate driver **10** of the present embodiment includes the transformer **40A** including the first coil **41A** and the second coil **42A** and the transformer **40B** including the first coil **41B** and the second coil **42B**. The gate driver **10** of the present embodiment further includes the capacitor **50A** connected between the ground terminal of the first coil **41A** and the ground terminal of the second coil **42A** and the capacitor **50B** connected between the ground terminal of the first coil **41B** and the ground terminal of the second coil **42B**.

(227) The first capacitor electrode **51A** of the capacitor **50A** and the first coil **41A** are electrically connected and have the same potential. The second capacitor electrode **52A** of the capacitor **50A** and the second coil **42A** are electrically connected and have the same potential. The first capacitor electrode **51B** of the capacitor **50B** and the first coil **41B** are electrically connected and have the same potential. The second capacitor electrode **52B** of the capacitor **50B** and the second coil **42B**

are electrically connected and have the same potential.

(228) The transformer **40A** and the transformer **40B** are used to, for example, transmit a set signal (SET) and a reset signal (RESET) from the low-voltage circuit **20** toward the high-voltage circuit **30**.

(229) At this time, the set signal output from the low-voltage circuit **20** cause a current $i_{\text{sub.1A}}$ to flow to the first coil **41A** of the transformer **40A**. A current $i_{\text{sub.2A}}$ flows to the second coil **42A**, which is magnetically coupled to the first coil **41A**. The current $i_{\text{sub.2A}}$ causes the high-voltage circuit **30** to generate a pulse signal, that is, receive a set signal.

(230) The set signal causes a current $i_{\text{sub.CA}}$ to flow between the first capacitor electrode **51A** and the second capacitor electrode **52A** of the capacitor **50A**. The current $i_{\text{sub.CA}}$ flows from the second capacitor electrode **52A** to the ground terminal of the second coil **42A**, namely, the ground **GND2**. This reduces the effect of the current $i_{\text{sub.CA}}$ flowing to the capacitor **50A** on the current $i_{\text{sub.2A}}$ flowing to the second coil **42A** of the transformer **40A**. Thus, the effect on signal transmission between the low-voltage circuit **20** and the high-voltage circuit **30** is reduced.

(231) The reset signal output from the low-voltage circuit **20** causes a current $i_{\text{sub.1B}}$ to flow to the first coil **41B** of the transformer **40B**. A current $i_{\text{sub.2B}}$ flows to the second coil **42B**, which is magnetically coupled to the first coil **41B**. The current $i_{\text{sub.2B}}$ causes the high-voltage circuit **30** to generate a pulse signal, that is, receive a reset signal.

(232) The reset signal causes a current $i_{\text{sub.CB}}$ to flow between the first capacitor electrode **51B** and the second capacitor electrode **52B** of the capacitor **50B**. The current $i_{\text{sub.CB}}$ flows from the second capacitor electrode **52B** to the ground terminal of the second coil **42B**, namely, the ground **GND2**. This reduces the effect of the current $i_{\text{sub.2B}}$ flowing to the second coil **42B** of the transformer **40B** on the current $i_{\text{sub.CB}}$ flowing to the capacitor **50B**. Thus, the effect on signal transmission between the low-voltage circuit **20** and the high-voltage circuit **30** is reduced.

(233) The phase of current $i_{\text{sub.CA}}$ is retarded from the phase of current i of the set signal in accordance with the impedance value of the transformer **40A** and the capacitance value of the capacitor **50A**. The phase of current $i_{\text{sub.CB}}$ is retarded from the phase of the current i of the reset signal in accordance with the impedance value of the transformer **40B** and the capacitance value of the capacitor **50B**. With this configuration, even when the currents $i_{\text{sub.CA}}$ and $i_{\text{sub.CB}}$ affect signal reception of the high-voltage circuit **30**, a mask period may be set in accordance with timings of the currents $i_{\text{sub.CA}}$ and $i_{\text{sub.CB}}$. The mask period is shorter than that for the case of the parasitic capacitance **C1**. This reduces the effect on the fast signal transmission.

Advantages

(234) The gate driver **10** of the present embodiment obtains the following advantages.

(235) (1-1) The transformer chip **80** includes the insulation layer **84**, the transformer **40A**, and the capacitor **50A**. The transformer **40A** includes the first coil **41A** and the second coil **42A** embedded in the insulation layer **84** and separated from each other in the thickness-wise direction of the insulation layer **84**. The capacitor **50A** includes the first capacitor electrode **51A** and the second capacitor electrode **52A** arranged between the first coil **41A** and the second coil **42A** of the transformer **40A**. The first capacitor electrode **51A** is connected to the first ground terminal **45** of the first coil **41A**. The second capacitor electrode **52A** is connected to the second ground terminal **48** of the second coil **42A**. The insulation layer **84** includes the insulation films **85** (**851** to **856**). The first capacitor electrode **51A** is formed between the second insulation film **85B** of the insulation film **851** and the first insulation film **85A** of the insulation film **852**. The second capacitor electrode **52A** is formed between the second insulation film **85B** of the insulation film **853** and the first insulation film **85A** of the insulation film **854**.

(236) With this structure, the transformer **40A** is used to, for example, transmit a set signal (SET) from the low-voltage circuit **20** toward the high-voltage circuit **30**. The set signal causes the current $i_{\text{sub.CA}}$ flowing through the capacitor **50A** to flow from the second capacitor electrode **52A** to the ground terminal of the second coil **42A**, namely, the ground **GND2**. This reduces the effect of the

current $i_{\text{sub.CA}}$ flowing to the capacitor **50A** on the current $i_{\text{sub.2A}}$ flowing to the second coil **42A** of the transformer **40A**. Thus, the effect on signal transmission between the low-voltage circuit **20** and the high-voltage circuit **30** is reduced.

(237) The transformer chip **80** includes the insulation layer **84**, the transformer **40B**, and the capacitor **50B**. The transformer **40B** includes the first coil **41B** and the second coil **42B** embedded in the insulation layer **84** and separated from each other in the thickness-wise direction of the insulation layer **84**. The capacitor **50B** includes the first capacitor electrode **51B** and the second capacitor electrode **52B** arranged between the first coil **41B** and the second coil **42B** of the transformer **40B**. The first capacitor electrode **51B** is connected to the first ground terminal **45** of the first coil **41B**. The second capacitor electrode **52B** is connected to the second ground terminal **48** of the second coil **42B**.

(238) The transformer **40B** is used to, for example, transmit a reset signal (RESET) from the low-voltage circuit **20** toward the high-voltage circuit **30**. The reset signal causes the current $i_{\text{sub.CB}}$ flowing through the capacitor **50B** to flow from the second capacitor electrode **52B** to the ground terminal of the second coil **42B**, namely, the ground GND2. This reduces the effect of the current $i_{\text{sub.2B}}$ flowing to the second coil **42B** of the transformer **40B** on the current $i_{\text{sub.CB}}$ flowing to the capacitor **50B**. Thus, the effect on signal transmission between the low-voltage circuit **20** and the high-voltage circuit **30** is reduced.

(239) (1-2) The phase of current $i_{\text{sub.CA}}$ flowing through the capacitor **50A** is retarded from the phase of current i of the set signal in accordance with the impedance value of the transformer **40A** and the capacitance value of the capacitor **50A**. The phase of current $i_{\text{sub.CB}}$ flowing through the capacitor **50B** is retarded from the phase of current i of the reset signal in accordance with the impedance value of the transformer **40B** and the capacitance value of the capacitor **50B**. With this configuration, even when the currents $i_{\text{sub.CA}}$ and $i_{\text{sub.CB}}$ affect signal reception of the high-voltage circuit **30**, a mask period may be set in accordance with phases of the currents $i_{\text{sub.CA}}$ and $i_{\text{sub.CB}}$. This simplifies the design of the high-voltage circuit **30**. The mask period is shorter than that for the case of the parasitic capacitance **C1**. This reduces the effect on the fast signal transmission.

(240) (1-3) In plan view, the dummy pattern **120** is arranged around the second coils **42A** and **42B**. This structure limits concentration of an electric field on the second coils **42A** and **42B**.

(241) (1-4) In plan view, the dummy pattern **125** is arranged around the second capacitor electrodes **52A** and **52B**. This structure limits concentration of an electric field on the second capacitor electrodes **52A** and **52B**.

(242) (1-5) The gate driver **10** includes the low-voltage circuit **20**, the high-voltage circuit **30**, and the transformer chip **80**. The low-voltage circuit **20** and the high-voltage circuit **30** are connected by the transformer chip **80** and configured to transmit a signal through the transformer chip **80**. The transformer chip **80** includes the transformers **40A** and **40B** and the capacitors **50A** and **50B**. The capacitor **50A** includes the first capacitor electrode **51A** and the second capacitor electrode **52A** arranged between the first coil **41A** and the second coil **42A** of the transformer **40A**. The capacitor **50B** includes the first capacitor electrode **51B** and the second capacitor electrode **52B** arranged between the first coil **41B** and the second coil **42B** of the transformer **40B**. The first capacitor electrodes **51A** and **51B** are connected to the first ground terminals **45** of the first coils **41A** and **41B**. The second capacitor electrodes **52A** and **52B** are connected to the second ground terminals **48** of the second coils **42A** and **42B**. This structure obtains the advantage (1) described above. Thus, the effect of the gate driver **10** on a transmitting signal is reduced.

(243) (1-6) In an example of a structure in which the gate driver **10** includes a transformer **40** and a capacitor **50**, the gate driver **10** may include a low-voltage circuit chip including the low-voltage circuit **20**, the transformer **40**, and the capacitor **50**. Alternatively, the gate driver **10** may include a high-voltage circuit chip including the high-voltage circuit **30**, the transformer **40**, and the capacitor **50**. In such structures, when the circuit configuration of the low-voltage circuit **20** or the high-

voltage circuit **30** is changed, the corresponding chip entirely needs to be changed. This increases costs when different types of gate drivers are manufactured.

(244) In this regard, in the present embodiment, the transformer **40** and the capacitor **50** are included in the transformer chip **80**, which is independent of the low-voltage circuit chip **60** and the high-voltage circuit chip **70**. That is, the chip is dedicated to the transformer **40**. This allows the common transformer chip **80** to be used for the low-voltage circuit **20** and the high-voltage circuit **30**, which differ from each other. As a result, costs are reduced when manufacturing different types of gate drivers **10** that differ in at least one of the low-voltage circuit **20** and the high-voltage circuit **30**.

(245) (1-7) The first coil **41A** and the first capacitor electrode **51A** of the present embodiment are located at opposite sides of the first insulation film **85A** of the insulation film **852**. The first coil **41A** and the first capacitor electrode **51A** are formed from a conductive metal. The first insulation film **85A** is formed from, for example, SiN. In other words, the first coil **41A**, the first insulation film **85A**, and the first capacitor electrode **51A** may have a metal-insulator-metal (MIM) structure. Also, the second coil **42A**, the second capacitor electrode **52A**, and the first insulation film **85A** of the insulation film **854** may have a metal-insulator-metal (MIM) structure. Thus, a capacitor having the MIM structure is readily formed on the transformer chip **80**.

(246) (1-8) The first capacitor electrode **51A** and the second capacitor electrode **52A** are formed from a nonmagnetic material. When, for example, TiN, CrSi, or the like is selected as a magnetic material, a resistive element may be readily formed from the nonmagnetic material on the transformer chip **80**.

Second Embodiment

(247) A second embodiment of a gate driver **10** will be described with reference to FIGS. **14**, **15A**, and **15B**. The gate driver **10** of the present embodiment differs from the gate driver **10** of the first embodiment mainly in the positions where the first capacitor electrode **51A** of the capacitor **50A** and the second capacitor electrode **52A** are formed. In the following description, the differences from the gate driver **10** of the first embodiment will be described in detail. The same reference characters are given to those components that are the same as the corresponding components of the gate driver **10** of the first embodiment. Such components will not be described in detail.

(248) As shown in FIG. **14**, the first capacitor electrode **51A** and the second capacitor electrode **52A** of the capacitor **50A** are arranged between the first coil **41A** and the second coil **42A** of the transformer **40A**. The first capacitor electrode **51A** and the second capacitor electrode **52A** are opposed to each other in the z-direction with multiple insulation films **85** interposed.

(249) At least one insulation film **85** is interposed between the first capacitor electrode **51A** and the first coil **41A**. In the present embodiment, one insulation film **85** is interposed. The second capacitor electrode **52A** is formed in the insulation film **853** located next to the insulation film **854**, in which the second coil **42A** is formed.

(250) As shown in FIGS. **14** and **15A**, the first capacitor electrode **51A** is configured to be a conductive layer embedded in one insulation film **85**. More specifically, the insulation film **852**, in which the first capacitor electrode **51A** is embedded, includes a first insulation film **85A** and a second insulation film **85B**. An insulation film **855** is formed between the insulation film **851**, including the first coil **41A**, and the insulation film **852** and includes a first insulation film **85A** and a second insulation film **85B**.

(251) The first capacitor electrode **51A** is formed on the upper surface of the insulation film **855**, more specifically, the upper surface of the second insulation film **85B** of the insulation film **855**. The first insulation film **85A** of the insulation film **852** is formed to cover the first capacitor electrode **51A**. Thus, the upper surface and the side surface of the first capacitor electrode **51A** are in contact with the first insulation film **85A** of the insulation film **852**. The lower surface of the first capacitor electrode **51A** is flush with the lower surface of the first insulation film **85A** of the insulation film **852**. The lower surface of the first capacitor electrode **51A** is in contact with the

second insulation film **85B** of the insulation film **855**. Thus, the first capacitor electrode **51A** is embedded in the insulation film **85**. The second insulation film **85B** of the insulation film **855** is arranged between the first capacitor electrode **51A** and the first coil **41A**. The first insulation film **85A** of the insulation film **852** corresponds to a first thin film. The second insulation film **85B** of the insulation film **855** corresponds to a first interlayer insulation film.

(252) As shown in FIGS. **14** and **15B**, the second capacitor electrode **52A** is configured to be a conductive layer embedded in one insulation film **85**. More specifically, the insulation film **853**, in which the second capacitor electrode **52A** is embedded, includes a first insulation film **85A** and a second insulation film **85B**. An insulation film **856** is located next to the insulation film **853** in a direction toward the first capacitor electrode **51A** and includes a first insulation film **85A** and a second insulation film **85B**. The second capacitor electrode **52A** is formed on the upper surface of the insulation film **856**, more specifically, the upper surface of the second insulation film **85B** of the insulation film **856**. The first insulation film **85A** of the insulation film **853** is formed to cover the second capacitor electrode **52A**. Thus, the upper surface and the side surface of the second capacitor electrode **52A** are in contact with the first insulation film **85A** of the insulation film **853**. The lower surface of the second capacitor electrode **52A** is flush with the lower surface of the first insulation film **85A** of the insulation film **853**. The lower surface of the second capacitor electrode **52A** is in contact with the second insulation film **85B** of the insulation film **856**. Thus, the second capacitor electrode **52A** is embedded in the insulation films **85**. In other words, the second insulation film **85B** of the insulation film **856** and the first coil **41A** are located at opposite sides of the first capacitor electrode **51A**. The second insulation film **85B** of the insulation film **853** is arranged between the first capacitor electrode **51A** and the first coil **41A**. The first insulation film **85A** of the insulation film **853** corresponds to a second thin film. The second insulation film **85B** of the insulation film **856** corresponds to a second interlayer insulation film.

(253) The distance between the first capacitor electrode **51A** and the second capacitor electrode **52A** is determined by the thickness of the insulation film **85** and the number of the insulation films **85** stacked between the first capacitor electrode **51A** and the second capacitor electrode **52A**. The distance is set in accordance with the insulation voltage or electric field strength of the transformer chip **80**.

(254) The first capacitor electrode **51A** is formed from a material including, for example, Ta, TaN, Ti, or TiN. The first capacitor electrode **51A** and the barrier film **143**, forming the first coil **41A**, the interconnect **131A**, and the like, are formed from the same material. The first capacitor electrode **51A** and the barrier film **143**, which includes vias embedded in the insulation film **855** located next to the insulation film **852** including the first capacitor electrode **51A**, are formed at the same time.

(255) As shown in FIG. **14**, the first capacitor electrode **51A** is electrically connected to the first coil **41A**.

(256) The first coil **41A** includes the first coil wiring **43A**, the first signal terminal **44A**, and the first ground terminal **45**. The first capacitor electrode **51A** includes the first electrode wiring **53A**, the first capacitor end **54A**, and the first capacitor ground terminal **55**. The first electrode wiring **53A** overlaps the first coil wiring **43A** in the z-direction. The first capacitor end **54A** overlaps the first signal terminal **44A** in the z-direction. The first capacitor ground terminal **55** overlaps the first ground terminal **45** in the z-direction.

(257) The first capacitor ground terminal **55** of the first capacitor electrode **51A** is connected to the first ground terminal **45** of the first coil **41A**. The insulation film **855**, which is arranged between the first capacitor electrode **51A** and the first coil **41A**, includes vias **137** extending through the insulation film **855** in the z-direction. The vias **137** connect the first capacitor ground terminal **55** of the first capacitor electrode **51A** and the first ground terminal **45** of the first coil **41A**. Thus, the first capacitor electrode **51A** is connected to the first ground terminal **45** of the first coil **41A**.

(258) As shown in FIG. **14**, the second capacitor electrode **52A** is electrically connected to the second coil **42A**.

(259) The second coil **42A** includes the second coil wiring **46A**, the second signal terminal **47A**, and the second ground terminal **48**. The second capacitor electrode **52A** includes the second electrode wiring **56A**, the second capacitor end **57A**, and the second capacitor ground terminal **58**. The second electrode wiring **56A** overlaps the second coil wiring **46A** in the z-direction. The second capacitor end **57A** overlaps the second signal terminal **47A** in the z-direction. The second capacitor ground terminal **58** overlaps the second ground terminal **48** in the z-direction.

(260) The second capacitor ground terminal **58** of the second capacitor electrode **52A** is connected to the second ground terminal **48** of the second coil **42A**. The insulation film **853**, which is arranged between the second capacitor electrode **52A** and the second coil **42A**, includes vias **138** extending through the insulation film **853** in the z-direction. The vias **138** further extend through the first insulation film **85A** of the insulation film **854**, in which the second coil **42A** is embedded. The vias **138** connect the second capacitor ground terminal **58** of the second capacitor electrode **52A** and the second ground terminal **48** of the second coil **42A**. The second capacitor electrode **52A** is connected to the second ground terminal **48** of the second coil **42A**.

(261) FIGS. **16A** and **16B** show some of manufacturing processes of the transformer chip **80** including processes of forming vias **131V** embedded in the insulation film **855** and forming the first capacitor electrode **51A**. The vias **131V** are included in the interconnect **131C**, shown in FIG. **14**.

(262) As shown in FIG. **16A**, via holes **151** are formed in the insulation film **855** and extends through the insulation film **855** in the thickness-wise direction. A barrier film **152** is formed on the surface of the insulation film **855**, more specifically, the upper surface of the second insulation film **85B**, and the side and bottom wall surfaces of the via holes **151**. The via holes **151** are filled with the body layer **144** at an inner side of the barrier film **152**. A resist film **153** is formed to cover predetermined portions of the upper surface of the barrier film **152**. The predetermined portions covered by the resist film **153** correspond to where the first capacitor electrode **51A** (refer to FIG. **14**) is formed. The barrier film **152** exposed from the resist film **153** (refer to FIG. **16B**) is removed by, for example, etching, and then the resist film **153** is removed. This obtains the first capacitor electrode **51A** shown in FIG. **16B**.

(263) Although not shown and described, the second capacitor electrode **52A** of the present embodiment may be formed through the same processes as the first capacitor electrode.

(264) Advantages

(265) The gate driver **10** of the present embodiment obtains the following advantages in addition to the advantages of the first embodiment.

(266) (2-1) The first capacitor electrode **51A** and the barrier film **143**, forming the first coil **41A**, the interconnect **131A**, and the like, are formed from the same material. This allows the first capacitor electrode **51A** to be formed at the same time as, for example, the barrier film **143** forming the vias **131V** included in the interconnect **131C**. Thus, the number of processes for the first capacitor electrode **51A** is not increased. The second capacitor electrode **52A** may also be formed through the same processes as the first capacitor electrode **51A**. Thus, the number of processes is not increased.

(267) (2-2) The first capacitor electrode **51A** and the barrier film **143**, forming the first coil **41A**, the interconnect **131A**, and the like, are formed from the same material. Thus, the barrier film **143** may be used to readily form other elements such as a resistive element on the transformer chip **80**.

Modified Examples

(268) The above embodiments exemplify, without any intention to limit, applicable forms of an insulation module and a gate driver according to the present disclosure. The insulation module and the gate driver according to the present disclosure can be applicable to forms differing from the above embodiments. In an example of such a form, the structure of the above embodiment is partially replaced, changed, or omitted, or a further structure is added to the above embodiment. The modified examples described below may be combined with one another as long as there is no

technical inconsistency. In the modified examples, the same reference characters are given to those components that are the same as the corresponding components of the above embodiment. Such components will not be described in detail.

(269) In the embodiments, the shape of the capacitor **50** in plan view may be changed.

(270) FIG. **17** shows a modified example of first capacitor electrodes **51A** and **51B**. The first capacitor electrode **51A** is elliptical and annular and overlaps the first coil wiring **43A** of the first coil **41A** shown in FIG. **5**. The first capacitor electrode **51A** of this modified example includes a first electrode wiring **53A** formed of annular wires arranged from an inner end **43i** of the first coil wiring **43A** to an outer end **43o** of the first coil wiring **43A**, which are shown in FIG. **5**. The first capacitor electrode **51A** includes a slit **51As** extending outward from the center. The first capacitor electrode **51A** is annular and open by the slit **51As**.

(271) The first capacitor electrode **51B** is elliptical and annular and overlaps the first coil wiring **43A** of the first coil **41B** shown in FIG. **5**. The first capacitor electrode **51B** of this modified example is formed of annular wires arranged from the inner end **43i** of the first coil wiring **43B** to the outer end **43o** of the first coil wiring **43B**, which are shown in FIG. **5**. The first capacitor electrode **51B** includes a first slit **51Bs** extending outward from the center. The first capacitor electrode **51B** is annular and open by the first slit **51Bs**.

(272) FIG. **18** shows a modified example of first capacitor electrodes **51A** and **51B**. The first capacitor electrode **51A** is elliptical and annular and overlaps the first coil wiring **43A** of the first coil **41A** shown in FIG. **5**. The first capacitor electrode **51A** of this modified example includes a first electrode wiring **53A** that is plate-shaped and continuous from the inner end **43i** of the first coil wiring **43A** to the outer end **43o** of the first coil wiring **43B**, which are shown in FIG. **5**. The first capacitor electrode **51A** includes a first slit **51As** extending outward from the center. The first capacitor electrode **51A** is annular and open by the first slit **51As**.

(273) The first capacitor electrode **51B** is elliptical and annular and overlaps the first coil wiring **43A** of the first coil **41B** shown in FIG. **5**. The first capacitor electrode **51B** of this modified example includes a first electrode wiring **53B** that is plate-shaped and continuous from the inner end **43i** of the first coil wiring **43A** to the outer end **43o** of the first coil wiring **43B**, which are shown in FIG. **5**. The first capacitor electrode **51B** includes a first slit **51Bs** extending outward from the center. The first capacitor electrode **51B** is annular and open by the first slit **51Bs**.

(274) FIG. **19** shows a modified example of first capacitor electrodes **51A** and **51B**. The first capacitor electrode **51A** is elliptical-plate-shaped and overlaps the first coil wiring **43A** and the first signal terminal **44A** of the first coil **41A**, shown in FIG. **5**. The first capacitor electrode **51A** of this modified example includes a first electrode wiring **53A** that is plate-shaped and continuous from the center of the first coil **41A** to the outer end **43o** of the first coil wiring **43B**, shown in FIG. **5**. Thus, the first electrode wiring **53A** of this modified example includes the first signal terminal **44A**, shown in FIG. **5**. The first capacitor electrode **51A** includes a first slit **51As** extending outward from the center.

(275) The first capacitor electrode **51B** is elliptical-plate-shaped and overlaps the first coil wiring **43A** and the first signal terminal **44B** of the first coil **41B**, shown in FIG. **5**. The first capacitor electrode **51B** of this modified example is plate-shaped and continuous from the center of the first coil **41B** to the outer end **43o** of the first coil wiring **43B**, shown in FIG. **5**. Thus, the first electrode wiring **53B** of this modified example includes the first signal terminal **44B**, shown in FIG. **5**. The first capacitor electrode **51B** includes a first slit **51Bs** extending outward from the center.

(276) In the embodiments, the cross section of the first coil **41A** and the second coil **42A**, the first capacitor electrode **51A** and the second capacitor electrode **52A**, and the dummy patterns **120** and **125** in the z-direction may be changed.

(277) FIGS. **20** to **28** schematically show portions of the transformer chip **80** in a cross section in the z-direction.

(278) As shown in FIG. **20**, the dummy pattern **120** corresponding to the second coil **42A** and the

dummy pattern **125** (refer to FIG. **9**) corresponding to the second capacitor electrode **52A** may be omitted.

(279) As shown in FIG. **21**, the dummy pattern **125** (refer to FIG. **9**) corresponding to the second capacitor electrode **52A** may be omitted so that only the dummy pattern **120** corresponding to the second coil wiring **46A** of the second coil **42A** is included. Alternatively, the dummy pattern **120** corresponding to the second coil wiring **46A** of the second coil **42A** may be omitted so that only the dummy pattern **125** (refer to FIG. **9**) corresponding to the second capacitor electrode **52A** is included.

(280) As shown in FIG. **22**, the first electrode wiring **53A** of the first capacitor electrode **51A** includes an outer end **53o** located at an outer side of the outer end **43o** of the first coil wiring **43A** of the first coil **41A**. The first electrode wiring **53A** of the first capacitor electrode **51A** includes an inner end **53i** located at the same position as the inner end **43i** of the first coil wiring **43A** of the first coil **41A**. The second electrode wiring **56A** of the second capacitor electrode **52A** includes an outer end **56o** located at an outer side of an outer end **46o** of the second coil wiring **46A** of the second coil **42A**. The second electrode wiring **56A** of the second capacitor electrode **52A** includes an inner end **56i** located at the same position as an inner end **46i** of the second coil wiring **46A** of the second coil **42A**.

(281) As shown in FIG. **23**, the outer end **53o** of the first electrode wiring **53A** of the first capacitor electrode **51A** is located at an outer side of the outer end **43o** of the first coil wiring **43A** of the first coil **41A**. The inner end **53i** of the first electrode wiring **53A** of the first capacitor electrode **51A** is located at an inner side of the inner end **43i** of the first coil wiring **43A** of the first coil **41A**. The outer end **56o** of the second electrode wiring **56A** of the second capacitor electrode **52A** is located at an outer side of the outer end **46o** of the second coil wiring **46A** of the second coil **42A**. The inner end **56i** of the second electrode wiring **56A** of the second capacitor electrode **52A** is located at an inner side of the inner end **46i** of the second coil wiring **46A** of the second coil **42A**.

(282) As shown in FIG. **24**, the first electrode wiring **53A** of the first capacitor electrode **51A** and the second electrode wiring **56A** of the second capacitor electrode **52A** overlap the second coil wiring **46A** of the second coil **42A** and the dummy pattern **120**. More specifically, the outer end **53o** of the first electrode wiring **53A** of the first capacitor electrode **51A** is located at the same position as the outer end of the dummy pattern **120** formed at an outer side of the second coil wiring **46A** of the second coil **42A**. In the same manner, the outer end **56o** of the second electrode wiring **56A** of the second capacitor electrode **52A** is located at the same position as the outer end of the dummy pattern **120** formed at an outer side of the second coil wiring **46A** of the second coil **42A**. The first electrode wiring **53A** of the first capacitor electrode **51A** includes an inner end **53i** located at the same position as the inner end **43i** of the first coil wiring **43A** of the first coil **41A**. The second electrode wiring **56A** of the second capacitor electrode **52A** includes an inner end **56i** located at the same position as an inner end **46i** of the second coil wiring **46A** of the second coil **42A**.

(283) As shown in FIG. **25**, the first electrode wiring **53A** of the first capacitor electrode **51A** and the second electrode wiring **56A** of the second capacitor electrode **52A** overlap the second coil wiring **46A** of the second coil **42A** and the dummy pattern **120**. More specifically, the outer end **53o** of the first electrode wiring **53A** of the first capacitor electrode **51A** is located at the same position as the outer end of the dummy pattern **120** formed at an outer side of the second coil wiring **46A** of the second coil **42A**. In the same manner, the outer end **56o** of the second electrode wiring **56A** of the second capacitor electrode **52A** is located at the same position as the outer end of the dummy pattern **120** formed at an outer side of the second coil wiring **46A** of the second coil **42A**. The inner end **53i** of the first electrode wiring **53A** of the first capacitor electrode **51A** is located at an inner side of the inner end **43i** of the first coil wiring **43A** of the first coil **41A**. The inner end **56i** of the second electrode wiring **56A** of the second capacitor electrode **52A** is located at an inner side of the inner end **46i** of the second coil wiring **46A** of the second coil **42A**.

(284) As shown in FIG. 26, the outer end 53o of the first electrode wiring 53A of the first capacitor electrode 51A is located at an inner side of the outer end 43o of the first coil wiring 43A of the first coil 41A. The inner end 53i of the first electrode wiring 53A of the first capacitor electrode 51A is located at an outer side of the inner end 43i of the first coil wiring 43A of the first coil 41A. The outer end 56o of the second electrode wiring 56A of the second capacitor electrode 52A is located at an inner side of the outer end 46o of the second coil wiring 46A of the second coil 42A. The inner end 56i of the second electrode wiring 56A of the second capacitor electrode 52A is located at an outer side of the inner end 46i of the second coil wiring 46A of the second coil 42A.

(285) As shown in FIG. 27, the width of the first electrode wiring 53A of the first capacitor electrode 51A is set to be less than the width of the first coil wiring 43A of the first coil 41A. That is, the ratio of wiring width to wiring spacing of the first electrode wiring 53A is set to be less than the ratio of wiring width to wiring spacing of the first coil wiring 43A. The width of the second electrode wiring 56A of the second capacitor electrode 52A is set to be less than the width of the second coil wiring 46A of the second coil 42A. That is, the ratio of wiring width to wiring spacing of the second electrode wiring 56A is set to be less than the ratio of wiring width to wiring spacing of the second coil wiring 46A.

(286) As shown in FIG. 28, the width of the first electrode wiring 53A of the first capacitor electrode 51A is set to be greater than the width of the first coil wiring 43A of the first coil 41A. That is, the ratio of wiring width to wiring spacing of the first electrode wiring 53A is set to be greater than the ratio of wiring width to wiring spacing of the first coil wiring 43A. The width of the second electrode wiring 56A of the second capacitor electrode 52A is set to be greater than the width of the second coil wiring 46A of the second coil 42A. That is, the ratio of wiring width to wiring spacing of the second electrode wiring 56A is set to be greater than the ratio of wiring width to wiring spacing of the second coil wiring 46A.

(287) As described above, the shape of the first coil wiring 43A (first coil 41A) and the second coil wiring 46A (second coil 42A) may be set in any manner. The shape of the first electrode wiring 53A (first capacitor electrode 51A) and the second electrode wiring 56A (second capacitor electrode 52A) may be set in any manner. In an example, in the z-direction, the thickness of the first coil wiring 43A and the second coil wiring 46A may differ from the thickness of the first electrode wiring 53A and the second electrode wiring 56A.

(288) In the embodiments, the passivation film 160 is not limited to a material including silicon nitride and may be any layer that protects the insulation layer 84.

(289) In the embodiments, the arrangement of the transformers 40A and 40B and the capacitors 50A and 50B may be changed in any manner. In an example, the transformer 40A, the transformer 40A, the transformer 40B, and the transformer 40B may be arranged in this order from the chip side surface 80c toward the chip side surface 80d of the transformer chip 80. The capacitors 50A and 50B are arranged in accordance with the positions of the transformers 40A and 40B.

(290) In the embodiments, the first dummy pattern 121 of the dummy pattern 120 is electrically connected to the second coil 42B. However, there is no limit to such a configuration. In an example, the first dummy pattern 121 may be arranged independently of the second coils 42A and 42B. In other words, the first dummy pattern 121 may be configured not to be electrically connected to the second coils 42A and 42B. In the embodiments, the third dummy pattern 123 is electrically connected to the first dummy pattern 121. However, there is no limit to such a configuration. In an example, the third dummy pattern 123 may be configured not to be electrically connected to the first dummy pattern 121.

(291) In the embodiments, the first dummy pattern 126 of the dummy pattern 125 is electrically connected to the first capacitor electrode 51A. However, there is no limit to such a configuration. In an example, the first dummy pattern 126 may be arranged independently of the first capacitor electrodes 51A and 51B. That is, the first dummy pattern 126 may be configured not to be electrically connected to the first capacitor electrodes 51A and 51B. In the embodiments, the third

dummy pattern **128** is electrically connected to the first dummy pattern **126**. However, there is no limit to such a configuration. In an example, the third dummy pattern **128** may be configured not to be electrically connected to the first dummy pattern **126**.

(292) In the embodiments, the structure of the dummy pattern **120** corresponding to the second coils **42A** and **42B** may be changed in any manner. In an example, one or two of the first dummy pattern **121**, the second dummy pattern **122**, and the third dummy pattern **123** may be omitted from the dummy pattern **120**. The dummy pattern **120** may be omitted from the transformer chip **80**.

(293) In the embodiments, the structure of the dummy pattern **125** corresponding to the second capacitor electrodes **52A** and **52B** may be changed in any manner. In an example, one or two of the first dummy pattern **126**, the second dummy pattern **127**, and the third dummy pattern **128** may be omitted from the dummy pattern **125**. The dummy pattern **125** may be omitted from the transformer chip **80**.

(294) In the embodiments, the low-voltage circuit **20** and the transformer **40** are mounted on separate chips. However, there is no limit to such a configuration. The transformer **40** and the low-voltage circuit **20** may be mounted on a single chip. In an example, the low-voltage circuit **20** may be formed on the substrate **83** of the transformer chip **80**. The transformer chip **80** is covered by the mold resin **110**.

(295) In the embodiments, the high-voltage circuit **30** and the transformer **40** are mounted on separate chips. However, there is no limit to such a configuration. The transformer **40** and the high-voltage circuit **30** may be mounted on a single chip. In an example, the high-voltage circuit **30** may be formed on the substrate **83** of the transformer chip **80**. In this case, the transformer chip **80** is mounted on the high-voltage die pad **101**. The transformer chip **80** is covered by the mold resin **110**.

(296) In the embodiments, the gate driver **10** may include an insulation module in which the transformers **40** are accommodated in a single package. The insulation module includes the transformer chip **80** and the mold resin **110** encapsulating the transformer chip **80**. The insulation module may further include a die pad on which the transformer chip **80** is mounted, leads, and wires connecting the leads to the transformer chip **80**. The mold resin **110** encapsulates the transformer chip **80**, the die pad, and the wires. The leads are configured to be electrically connected to both the low-voltage circuit **20** and the high-voltage circuit **30**.

(297) In the embodiments, the gate driver **10** may include a low-voltage circuit unit in which the low-voltage circuit **20** and the transformer **40** are accommodated in a single package. The low-voltage circuit unit may include the low-voltage circuit chip **60**, the transformer chip **80**, and the mold resin **110** encapsulating the low-voltage circuit chip **60** and the transformer chip **80**. The low-voltage circuit unit may further include a die pad, first leads, first wires connecting the first leads to the low-voltage circuit chip **60**, second leads, and second wires connecting the second leads to the transformer chip **80**. The mold resin **110** encapsulates at least the low-voltage circuit chip **60**, the transformer chip **80**, the die pad, and the wires. In an example, the first leads are configured to be electrically connected to the ECU **503**, and the second leads are configured to be electrically connected to the high-voltage circuit **30**.

(298) In the embodiments, the gate driver **10** may include a high-voltage circuit unit in which the high-voltage circuit **30** and the transformer **40** are accommodated in a single package. The high-voltage circuit unit may include the high-voltage circuit chip **70**, the transformer chip **80**, and the mold resin **110** encapsulating the high-voltage circuit chip **70** and the transformer chip **80**. The high-voltage circuit unit may further include a die pad, first leads, first wires connecting the first leads to the high-voltage circuit chip **70**, second leads, and second wires connecting the second leads to the transformer chip **80**. The mold resin **110** encapsulates at least the high-voltage circuit chip **70**, the transformer chip **80**, the die pad, and the wires. In an example, the first leads are configured to be electrically connected to the source of the switching element **501**, and the second leads are configured to be electrically connected to the low-voltage circuit **20**.

(299) In the embodiments, the high-voltage circuit **30** may be configured to transmit a signal to the low-voltage circuit **20** through the transformer **40** and the capacitor **50**. Alternatively, the low-voltage circuit **20** and the high-voltage circuit **30** may be configured to bidirectionally transmit signals to each other through the transformer **40** and the capacitor **50**.

(300) In the embodiments, the first coil **41A** and the second coil **42A** may have different number of windings, and the first coil **41B** and the second coil **42B** may have different number of windings. The first coil **41A** and the second coil **42A** may be wound in different directions, and the first coil **41B** and the second coil **42B** may be wound in different directions.

(301) In the embodiments, the positions of the slits **51As** and **51Bs** shown in FIG. 7 may be changed in any manner. In FIG. 7, the slits **51As** and **51Bs** extend in the x-direction. Instead, the slits **51As** and **51Bs** may extend in the y-direction. The slits **51As** and **51Bs** may extend in the same direction, for example, a direction toward the chip side surface **80c**. Also, the positions of the slits **52As** and **52Bs** shown in FIG. 8 may be changed. The slit **51As** shown in FIG. 7 and the slit **52As** shown in FIG. 8 may extend in different directions. Also, the slit **51Bs** shown in FIG. 7 and the slit **52Bs** shown in FIG. 8 may extend in different directions.

(302) In the present disclosure, the term “on” includes the meaning of “above” in addition to the meaning of “on” unless otherwise clearly indicated in the context. Thus, the phrase “A is formed on B” is intended to mean that A may be disposed directly on B in contact with B in the present embodiment and also that A may be disposed above B without contacting B in a modified example. In other words, the term “on” does not exclude a structure in which another member is formed between A and B.

(303) The z-direction as referred to in the present disclosure does not necessarily have to be the vertical direction and does not necessarily have to fully conform to the vertical direction. In the structures according to the present disclosure, “upward” and “downward” in the z-direction as referred to in the present description are not limited to “upward” and “downward” in the vertical direction. In an example, the x-direction may conform to the vertical direction. In another example, the y-direction may conform to the vertical direction.

(304) In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

REFERENCE SIGNS LIST

(305) **10**) gate driver **10R**) gate driver **20**) low-voltage circuit **21A**, **21B**) low-voltage signal line **30**) high-voltage circuit **31A**, **31B**) high-voltage signal line **40**) transformer **40A**, **40B**) transformer **41A**, **41B**) first coil **42A**, **42B**) second coil **43A**, **43B**) first coil wiring **43i**) inner end **43o**) outer end **44A**, **44B**) first signal terminal **45**) first ground terminal **46A**, **46B**) second coil wiring **46i**) inner end **46o**) outer end **47A**, **47B**) second signal terminal **48**) second ground terminal **50**) capacitor **50A**, **50B**) capacitor **51A**, **51B**) first capacitor electrode **51As**, **51Bs**) first slit **52A**, **52B**) second capacitor electrode **52As**, **52Bs**) second slit **53A**, **53B**) first electrode wiring **53i**) inner end **53o**) outer end **54A**, **54B**) first capacitor end **55**) first capacitor ground terminal **55A**, **55B**) interconnect **56A**, **56B**) second electrode wiring **56i**) inner end **56o**) outer end **57A**, **57B**) second capacitor end **58**) second capacitor ground terminal **58A**, **58B**) interconnect **60**) low-voltage circuit chip **60s**) chip main surface **61**) first electrode pad **62**) second electrode pad **63**) third electrode pad **70**) high-voltage circuit chip **70s**) chip main surface **71**) first electrode pad **72**) second electrode pad **73**) third electrode pad **80**) transformer chip **80a** to **80d**) chip side surface **80r**) chip back surface **80s**) chip main surface **81**) first electrode pad **81A** to **81C**) first electrode pad **82**) second electrode pad **82A** to **82C**) second electrode pad **83**) substrate **83r**) substrate back surface **83s**) substrate main surface **84**) insulation layer **84s**) surface **85**) insulation film **85A**) first insulation film **85B**) second insulation film **85B1**) insulation film **85B2**) insulation film **85L**) insulation film **85U**) insulation film **86**) shield electrode **87**) inner region **88**) outer region **89**) via **90**) low-voltage lead frame **91**) low-voltage die pad **92**) low-voltage lead **100**) high-voltage lead frame **101**) high-voltage die pad **102**) high-voltage lead **110**) mold resin **111** to **114**) resin side surface **120**) dummy

pattern 121) first dummy pattern 122) second dummy pattern 123) third dummy pattern 125) dummy pattern 126) first dummy pattern 127) second dummy pattern 128) third dummy pattern 131A to 131C) interconnect 131V) via 132A, 132C) first interconnect part 133A, 133C) second interconnect part 134A, 134C) via 135A, 135C) via 136 to 138) via 141, 142) coil groove 143) barrier film 144) body layer 145) first opening 145) opening 146) second opening 150) protection film 151) via hole 152) barrier film 153) resist film 160) passivation film 180) resin layer 181) inner resin layer 182) outer resin layer 183) separation groove 184) first resin opening 185) second resin opening 500) inverter device 501, 502) switching element 503) ECU 851 to 856) insulation film GND1, GND2) ground i) current i.sub.1A, i.sub.1B) current i.sub.2A, i.sub.2B) current i.sub.CA, i.sub.CB) current SD) conductive bonding material V1) first voltage V2) second voltage; W1 to W4) wire

Claims

1. An isolation transformer, comprising: an insulation layer; a transformer embedded in the insulation layer and including a first coil and a second coil separated from each other in a thickness-wise direction of the insulation layer, the first coil including a first signal terminal and a first ground terminal and being configured to receive a low voltage through the first signal terminal, and the second coil including a second signal terminal and a second ground terminal and being configured to receive a high voltage through the second signal terminal; and a capacitor including a first capacitor electrode and a second capacitor electrode, the first capacitor electrode being arranged between the first coil and the second coil and being connected to the first ground terminal, and the second capacitor electrode being arranged between the first capacitor electrode and the second coil and connected to the second ground terminal, wherein the insulation layer includes thin films and interlayer insulation films alternately stacked in the thickness-wise direction, the thin films include a first thin film and a second thin film separated from each other in the thickness-wise direction, the interlayer insulation films include a first interlayer insulation film located next to the first thin film in the thickness-wise direction and a second interlayer insulation film located next to the second thin film in the thickness-wise direction, the first capacitor electrode is formed between the first thin film and the first interlayer insulation film, and the second capacitor electrode is formed between the second thin film and the second interlayer insulation film.
2. The isolation transformer according to claim 1, wherein the first thin film is arranged between the first capacitor electrode and the first coil, and the second thin film is arranged between the second capacitor electrode and the second coil.
3. The isolation transformer according to claim 1, wherein the first interlayer insulation film is arranged between the first capacitor electrode and the first coil, the second interlayer insulation film and the second coil are located at opposite sides of the second capacitor electrode, and at least one of the interlayer insulation films is arranged between the second thin film and the second coil.
4. The isolation transformer according to claim 1, further comprising: a chip main surface and a chip back surface facing opposite directions in the thickness-wise direction, wherein the first coil is located closer to the chip back surface than the second coil is; a first ground electrode arranged on the chip main surface; and a first voltage interconnect connecting the first ground electrode to the first coil and the first capacitor electrode, wherein the first voltage interconnect includes a via including a barrier film arranged on a wall surface of a via hole formed in the insulation layer, and the barrier film of the via and at least one of the first capacitor electrode and the second capacitor electrode are formed from a same material.
5. The isolation transformer according to claim 1, wherein at least one of the first capacitor electrode and the second capacitor electrode includes a nonmagnetic material.
6. The isolation transformer according to claim 1, wherein as viewed in the thickness-wise

direction, the first capacitor electrode is annular and open and includes a first slit extending in a direction from a center of the first coil toward an outer side of the first coil, and as viewed in the thickness-wise direction, the second capacitor electrode is annular and open and includes a second slit extending in a direction from a center of the second coil toward an outer side of the second coil.

7. The isolation transformer according to claim 1, wherein the first coil includes a spiral first coil wiring, the first ground terminal connected to one end of the first coil wiring, and the first signal terminal connected to another end of the first coil wiring, and the second coil includes a spiral second coil wiring, the second ground terminal connected to one end of the second coil, and the second signal terminal connected to another end of the second coil wiring.

8. The isolation transformer according to claim 7, wherein as viewed in the thickness-wise direction, the first capacitor electrode includes a first electrode wiring overlapping the first coil wiring, a first capacitor end overlapping the first signal terminal, and a first capacitor ground terminal overlapping the first ground terminal, and as viewed in the thickness-wise direction, the second capacitor electrode includes a second electrode wiring overlapping the second coil wiring, a second capacitor end overlapping the second signal terminal, and a second capacitor ground terminal overlapping the second ground terminal.

9. The isolation transformer according to claim 8, wherein the first electrode wiring is set to be equal to the first coil wiring in ratio of wiring width to wiring spacing, and the second electrode wiring is set to be equal to the second coil wiring in ratio of wiring width to wiring spacing.

10. The isolation transformer according to claim 7, wherein the first capacitor electrode is plate-shaped and continuous from an inner end of the first coil wiring to an outer end of the first coil wiring as viewed in the thickness-wise direction, and the second capacitor electrode is plate-shaped and continuous from an inner end of the second coil wiring to an outer end of the second coil wiring as viewed in the thickness-wise direction.

11. The isolation transformer according to claim 8, wherein as viewed in the thickness-wise direction, an outer end of the second electrode wiring is located at either an outer side of an outer end of the second coil wiring or a same position as an outer end of the second coil wiring.

12. The isolation transformer according to claim 11, wherein as viewed in the thickness-wise direction, an inner end of the second electrode wiring is located at either a same position as an inner end of the second coil wiring or an inner side of an inner end of the second coil wiring.

13. The isolation transformer according to claim 11, wherein as viewed in the thickness-wise direction, an outer end of the first electrode wiring is located at either an outer side of an outer end of the first coil wiring or a same position as an outer end of the first coil wiring.

14. The isolation transformer according to claim 13, wherein as viewed in the thickness-wise direction, an inner end of the first electrode wiring is located at a same position as an inner end of the first coil wiring.

15. The isolation transformer according to claim 13, wherein as viewed in the thickness-wise direction, an inner end of the first electrode wiring is located at an inner side of an inner end of the first coil wiring.

16. The isolation transformer according to claim 4, further comprising: a dummy coil pattern arranged around the second coil and connected to the second coil.
